Local order parameters for ice polymorphs: a unified comparison and practical guide

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Abstract

Molecular simulations of water and ice rely on local order parameters to tell solid from liquid, one ice polymorph from another, and to measure nuclei and interfaces one molecule at a time. Dozens of such parameters exist, from tetrahedral order and Steinhardt invariants to CHILL+, template matching and machine-learned descriptors, but they are usually compared with values taken from different codes, cutoffs and water models. We review them in one taxonomy and present a reproducible benchmark that asks when they give reliable, transferable labels. Its central result concerns nucleus size. Near an ice/liquid interface every labeler misses a surface layer, and the number of molecules it misses per unit area, measured once on a planar interface (a dead layer of 3.4 Å for CHILL+ and 5.0 Å for PTM), predicts how strongly finite nuclei are undercounted. In replicated TIP4P/Ice seeds of 200–1600 molecules at 220 K, inverting the calibration recovers the constructed size from the labeled count within 5% pooled and 10% for any single seed with CHILL+ (10% and 25% with PTM), and from the largest labeled cluster within 10% from 400 molecules on. Fresh seeds of 300 and 600 molecules, prepared only after the calibration had been frozen, are recovered within 10% by both labelers. Without it, a labeler that is 99.7% accurate in bulk ice still counts only half of a 400-molecule nucleus, and two labelers can disagree on the size of the same nucleus by a factor of two. The broader benchmark (22 parameter families, 39 outputs and three labelers computed with one open implementation; Ih, Ic, stacking-disordered ice and supercooled liquid with five independent preparations, extended to TIP4P/2005, mW, six high-pressure ices with pressure-matched liquids, and 18 ideal structures) shows that locally averaged parameters separate ice from liquid and hexagonal from cubic ice almost perfectly, with thresholds that transfer between preparations, water models and neighbor cutoffs in the conditions tested, whereas distance-based parameters do not transfer and several popular rules form false nuclei of tens to hundreds of molecules in supercooled or compressed liquid. No single parameter separates every high-pressure ice from liquid at its own pressure, and none responds to proton order itself: proton order–disorder partners are separated only where their oxygen lattices differ, and Ih/XI and V/XIII not at all. A supplementary transfer test shows that the same parameters detect isolated confined water in unbinding paths through its missing water neighbors. We conclude with recommendations tied to the tested conditions, a reporting checklist and an archived, checksummed dataset.

Contents

1 Introduction **3**

2 The ice landscape in brief **4**
2.1 Phase diagram and stability 4
2.2 Structural signatures relevant to local order 5
2.3 Amorphous ices, clathrates and liquid water 6

3 A taxonomy of local order parameters **6**
3.1 Common conventions 7
3.2 Radial and density-based parameters 7
3.3 Angular and tetrahedral parameters 8
3.4 Hydrogen-bond and topology-based parameters 8
3.5 Bond-orientational order 8
3.6 Template and symmetry-based parameters 9
3.7 Other recent parameters 9
3.8 Summary table 9

4 Descriptors and machine-learned approaches **10**
4.1 High-dimensional descriptors 10
4.2 Supervised classifiers 10
4.3 Unsupervised approaches 11
4.4 Data-driven collective variables 11

5 A unified benchmark **12**
5.1 Datasets 12
5.2 Separability metric 14
5.3 Ideal crystals 15
5.4 Thermal motion: averaging is what survives 15
5.5 How far do thresholds transfer? 17
5.6 High-pressure phases 20
5.7 Polytype recall against depth in constructed seeds 22
5.8 False nuclei in liquid references 27
5.9 Application: nuclei along nucleation paths 29
5.10 Beyond ice: confined water 29
5.11 Sensitivity to the neighbor convention 29
5.12 Computational cost 31
5.13 What the benchmark does not cover 32

6 Practical guide **32**
6.1 Software 34
6.2 Reproducibility checklist 35

7 Outlook **35**

1 Introduction

Water freezes into more crystalline structures than almost any other simple substance. Ice Ih, the form of snow and glaciers, is joined by more than twenty crystalline phases that are stable or metastable at higher pressures, lower temperatures or both, by several amorphous ices, and by empty and filled clathrate frameworks [1, 2, 3]. New phases are still being found: ice XIX was characterized in 2021 [4, 5], a metastable ice XXI was obtained from supercompressed water at room temperature [6], and ice XXII has been reported above 300 GPa [7]. Molecular simulation is the main tool for asking how these phases form: how ice nucleates from supercooled water [8, 9, 10, 11, 12], which polymorph or stacking sequence grows [13, 14, 11], and how ice behaves at interfaces and in confinement [15, 16].

Each of these questions requires deciding, for every molecule in every frame, what kind of local environment it is in. That decision is made with a *local order parameter* (LOP): a function of the positions of a molecule and its neighbors that takes characteristic values in each phase. A LOP can do three different jobs. As a *descriptor* its distribution fingerprints a phase or a state point. As a *classifier*, with a threshold or a set of rules, it labels individual molecules, from which nuclei, interfaces and phase fractions are built. As a *collective variable* (CV) it drives enhanced sampling, which additionally requires it to be continuous and differentiable in the coordinates [17, 18]. Success at one job does not imply success at another. A parameter whose distributions differ on average may still misclassify many molecules, and a good classifier may be useless as a CV.

The choice matters. Computed nucleation barriers and rates from seeding depend on the order parameter used to measure the nucleus [19, 20, 21], and the size of a nucleus or the width of an interface changes with the labeling criterion [22, 15]. Popular water order parameters are only weakly correlated with one another and probe different distortions of the hydrogen-bond network [23, 24]. Yet most comparisons take values from different codes, with different neighbor definitions, cutoffs, water models and reference structures, so that the parameter cannot be separated from the way it was computed. Machine-learned classifiers report accuracies above 99% [25, 26, 16], but on their own datasets, which makes them hard to compare with each other or with simpler parameters. Common-setting comparisons do exist: Doi and co-workers screened 493 order parameters and their combinations for Ih, Ic and liquid at coexistence and for the high-pressure triple points III–V–liquid and V–VI–liquid [27, 28, 29], and Shi and co-workers compared labelers on the quasi-liquid layer of ice [15]. What is less well established is whether such rankings are reproducible and transferable: how much of a separation survives thermal motion rather than only energy minimization, whether a threshold fitted on one trajectory works on an independently prepared one or on another water model, how strongly results depend on cutoffs and averaging radii, and how close to an interface a labeler can still assign the right polytype. These are the questions our benchmark addresses.

The last question turns out to have a quantitative answer with direct consequences for nucleation studies. Every labeler misses a surface layer of a nucleus, so labeled nuclei are smaller than the crystals they belong to, by an amount that depends on the labeler and on the nucleus size. We show that the thickness of that layer, measured once on a planar ice/liquid interface, predicts the undercounting of finite seeds, and that inverting the relation recovers the size of a nucleus

from its labeled count alone, validated on independently prepared seeds that played no part in the calibration and confirmed on seeds prepared after it had been frozen (section 5.7). Nucleus sizes from seeding and enhanced-sampling studies are known to depend on the order parameter that defines them [19, 20, 21], and for hard spheres this dependence is captured by a constant shift of the nucleus radius fitted to the nucleation barrier [30]. The planar calibration measures such a shift independently, before any nucleus is seen, and so explains a spatial contribution to the dependence and offers a correction for it.

This review has four parts. Section 2 summarizes the ice landscape and the structural features a local parameter can and cannot see. Section 3 defines the physically motivated parameters in one notation, grouped by the information they use, and section 4 reviews descriptors, learned classifiers and data-driven collective variables. Section 5 reports a reproducible benchmark of selected local parameters for crystalline ice and supercooled water, in which every parameter is computed by one open implementation, SolOrder, with settings fixed before any comparison. It examines thermal robustness, threshold transferability, neighbor conventions and the spatial limits of local classification; table 3 lists exactly what it covers. Section 6 turns the results into recommendations and a reporting checklist, and section 7 lists open problems.

Scope. We consider per-molecule parameters for bulk crystalline ices, empty clathrate frameworks and liquid water, computed from oxygen positions or full molecular geometries. We do not cover global crystallographic analysis, spectroscopic observables, electronic structure or superionic phases. Amorphous ices, machine-learned classifiers and collective variables are reviewed from the literature but not benchmarked.

2 The ice landscape in brief

2.1 Phase diagram and stability

Figure 1 shows the experimental phase diagram. Roman numerals now run to XXII; counting Ih and Ic separately gives more than twenty crystalline phases [1, 2, 31, 6, 7]. Two caveats apply to any count. The structure of ice XIX is disputed: it was assigned to a partially antiferroelectric ordering of ice VI in a $\sqrt{2} \times \sqrt{2} \times 1$ supercell [4, 5], while an alternative analysis finds HCl-doped ice VI to remain disordered and distorted [32]. And the numeral XXII has been used for two different structures, a high-pressure orthorhombic phase [7] and, in a preprint, a metastable phase grown from emulsified water at 250 K [33]. Beyond the numbered phases, ice VII becomes plastic (freely rotating molecules on a bcc lattice) at high temperature, as predicted by simulation [34] and observed by quasi-elastic neutron scattering [35], and superionic phases form at planetary conditions [36, 37, 38, 39].

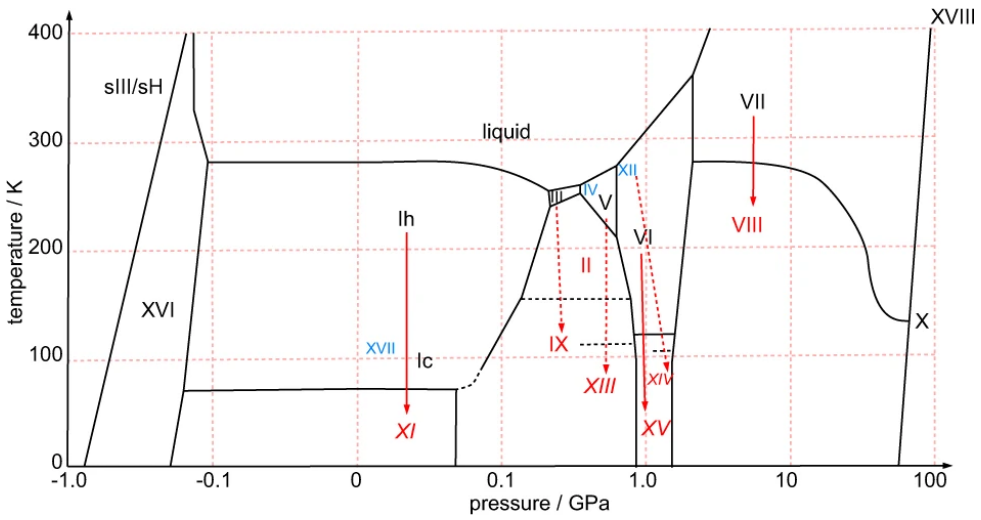


Figure 1: Experimental phase diagram of water. Black numerals: hydrogen-disordered phases; red: hydrogen-ordered; blue: metastable. Red arrows mark order–disorder transitions. Reproduced from Ref. [2] (CC BY 4.0).

Most phases come in pairs. Ih/XI, III/IX, V/XIII, VI/XV, VII/VIII and XII/XIV share an oxygen framework and differ in whether the protons are ordered; the ordered form is stable at lower temperature [40, 41, 42]. Stacking adds a second kind of variety: ice I can stack its puckered layers hexagonally (Ih), cubically (Ic) or in any mixture, and ice that freezes from supercooled water is stacking-disordered (Isd) rather than cubic [14, 43, 44]. Stacking-disorder-free Ic has been made only indirectly, by emptying ice XVII or a hydrogen hydrate [45, 46]. Simulated critical nuclei are themselves stacking-disordered [11, 10, 47].

Simulations see the phase diagram of the water model, not that of water. Rigid nonpolarizable models reproduce its topology to different degrees [48, 49, 50, 51]. TIP4P/Ice melts at 272.2 K [52] and its full phase diagram has been mapped with enhanced sampling [53]; TIP4P/2005 melts about 20 K lower [54, 55] and its high-pressure diagram includes a plastic phase [34, 56]; the oxygen-only mW model melts at 274.6 K [57]. Machine-learned potentials trained on first-principles data now reproduce the phase diagram semi-quantitatively, including nuclear quantum effects [58, 59, 60, 61, 62]. A state point that is stable for one model can therefore be metastable, or a different phase, for another, and every simulation study of ice structure should state where its state point lies on the model's own phase diagram.

2.2 Structural signatures relevant to local order

Table 1 lists, for the phases used in our benchmark, the quantities a local parameter could exploit, computed from ideal GenIce structures [63, 64]. Three groups stand out. The low-density, fully tetrahedral frameworks (Ih, Ic, XVII and the clathrates) have exactly four oxygens within 3.5 Å, near-perfect tetrahedral order Q and a large gap to the fifth neighbor ($d_5 \approx 4.3$–4.4 Å). The intermediate-pressure phases (II, III, V, XII) keep four hydrogen bonds but fold the network so that non-bonded oxygens enter the first shell: four to six neighbors within 3.5 Å, Q of 0.79–0.86 and d_5 of 3.2–3.5 Å. Ice VI and the VII/VIII family consist of two interpenetrating networks, giving 11–14 neighbors within 3.5 Å and Q as low as 0.33–0.69, even though every molecule still donates and accepts two hydrogen bonds.

Table 1: Structural signatures of the crystalline phases in the benchmark. Density, coordination number (CN, O within 3.5 Å), mean tetrahedral order Q and fifth-neighbor distance d_5 are computed with SolOrder on ideal GenIce structures (ice Ic and Ih are shown with the same lattice spacing). Proton order: D disordered, O ordered; partner: the phase sharing the oxygen framework.

Phase	Space group	H	Partner	ρ (g cm^{-3})	CN	Q	d_5 (Å)
Ih	$P6_3/mmc$	D	XI	0.92	4.0	0.998	4.43
Ic	$Fd\bar{3}m$	D	–	0.92	4.0	0.998	4.43
II	$R\bar{3}$	O	–	1.18	5.5	0.83	3.23
III	$P4_12_12$	D	IX	1.17	4.5	0.86	3.53
V	$C2/c$	D	XIII	1.24	5.8	0.80	3.32
VI	$P4_2/nmc$	D	XV	1.37	11.5	0.69	3.36
VII	$Pn\bar{3}m$	D	VIII	1.60	14.0	0.40	2.90
VIII	$I4_1/amd$	O	VII	1.63	13.0	0.33	2.88
XII	$I\bar{4}2d$	D	XIV	1.44	9.6	0.79	3.28
XVII	$P6_122$	D	–	0.88	4.0	0.97	4.29
sI (empty)	$Pm\bar{3}n$	D	–	0.80	4.0	0.98	4.40

What a local parameter cannot see follows from the same table. Proton-order partners share an oxygen sublattice, so any oxygen-only parameter can separate them only through the small lattice distortions that accompany ordering. Ih and Ic have identical first and nearly identical second shells; they differ in the relative orientation of neighboring tetrahedra (staggered versus eclipsed bonds) and in the third shell, so a parameter must either encode bond orientations beyond the first shell or average over neighbors. Rings are equally uninformative here: both Ih and Ic consist of six-membered rings only.

2.3 Amorphous ices, clathrates and liquid water

Low-density (LDA), high-density (HDA) and very-high-density amorphous ices are glassy solids whose local structure resembles that of liquids at similar density [65, 66, 67, 68], and a medium-density amorphous ice obtained by ball-milling Ih falls in the gap between LDA and HDA [69]. Clathrate hydrates build polyhedral cages from five- and six-membered rings around guest molecules; their empty frameworks (ices XVI and XVII among them) are tetrahedral like Ih [70, 71]. Supercooled liquid water, finally, becomes increasingly tetrahedral on cooling and has been described as a fluctuating mixture of low- and high-density local structures [72, 73, 74, 75]; it contains transient ice-like environments [76, 77]. These three are the hardest cases for local parameters, because they share the tetrahedral first shell of crystalline ice and differ from it mainly in medium-range order.

3 A taxonomy of local order parameters

We group physically motivated parameters by the information they use: distances (radial), angles within the first shell, the hydrogen-bond network, bond orientations expressed in spherical harmonics, and comparison with a template. Table 2 summarizes them with the settings used in the benchmark.

3.1 Common conventions

Throughout, $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$ is the minimum-image vector between oxygens i and j and r_{ij} its length. Every parameter needs a neighbor set $N_b(i)$, and the choice is part of the definition: a radial cutoff (3.5 Å for the first shell of ambient ices, near the first minimum of g_{OO}), a fixed number of nearest neighbors (four for tetrahedral parameters), Voronoi neighbors, or hydrogen-bond partners. These choices give different results [78], especially at high pressure, where non-bonded oxygens enter the first coordination shell. Many parameters exist in a *locally averaged* form,

$$\bar{X}(i) = \frac{1}{|N_b(i)| + 1} \sum_{j \in \{i\} \cup N_b(i)} X(j), \tag{1}$$

introduced for bond-orientational order by Lechner and Dellago [79] (who average the complex q_{lm} vectors rather than the scalar, section 3.5). Averaging trades spatial resolution for noise suppression, and as we will see it is the single most important ingredient at finite temperature. Finally, a parameter used as a CV must be a continuous, differentiable function of the coordinates. Hard cutoffs, nearest- neighbor ranks and integer counts are not; switching functions and Gaussian broadening help, but only if neighbor membership and averaging are smooth as well.

3.2 Radial and density-based parameters

The *coordination number* counts oxygens within r_c; its smooth form replaces the step function with a switching function,

$$\mathrm{CN}_s(i) = \sum_{j \neq i} \frac{1 - (r_{ij}/r_0)^n}{1 - (r_{ij}/r_0)^m}, \qquad n = 12,\ m = 28. \tag{2}$$

The *fifth-neighbor distance* d_5, the distance to the fifth-nearest oxygen, measures how far the second shell is from the first and is large in low-density, tetrahedral environments [80, 81]. The *local structure index* [82, 83] sorts the distances $r_1 < r_2 < \cdots < r_n < 3.7$ Å $< r_{n+1}$ and measures the variance of the gaps $\Delta_k = r_{k+1} - r_k$,

$$\mathrm{LSI}(i) = \frac{1}{n} \sum_{k=1}^{n} \left(\Delta_k - \bar{\Delta}\right)^2, \tag{3}$$

which is large when a clear gap separates the first and second shells. The Voronoi cell of an oxygen gives a local density $1/V$ and a shape measure, the asphericity $\eta = A^3/(36\pi V^2)$, which is 1 for a sphere and 2.25 for ice Ih [84, 85, 86]. The *pair-entropy fingerprint* [87, 88] projects the two-body excess entropy onto each particle,

$$s(i) = -2\pi\rho k_B \int_0^{r_m} \left[g_m^i(r) \ln g_m^i(r) - g_m^i(r) + 1\right] r^2\, dr, \tag{4}$$

where $g_m^i(r)$ is a Gaussian-broadened radial distribution centered on i and r_m is a cutoff of about 5 Å. Its local average $\bar{s}$ was designed as a crystallization CV and uses the whole radial structure out to the second shell.

3.3 Angular and tetrahedral parameters

The *orientational tetrahedral order* [89, 90],

$$Q(i) = 1 - \frac{3}{8}\sum_{j=1}^{3}\sum_{k=j+1}^{4}\left(\cos\psi_{jik} + \frac{1}{3}\right)^2, \tag{5}$$

uses the angles ψ_{jik} between the four nearest oxygens; it is 1 for a regular tetrahedron and 0 on average for an ideal gas. The *translational tetrahedral order* $S_k = 1 - \frac{1}{3}\sum_{k=1}^{4}(r_k - \bar{r})^2/(4\bar{r}^2)$ measures the variance of the four nearest distances [89]. A family of parameters developed for structure classification by Takahashi, Doi and co-workers [91, 27, 92] uses all angles θ_{jik} in the first shell. The bond-angle parameter averages $\cos^{n_1}(n_2\theta_{jik} + \phi)$ [93, 92]; the angular Fourier series parameter averages $f(\min(r_{ij}, r_{ik}))\cos(a\theta_{jik})$ [94, 92]; the angle-histogram parameters H_1–H_3 are Fourier amplitudes of the histogram of θ_{jik} [92]; and the common-neighborhood parameters A_1–A_3 sum vectors to neighbors shared by i and each of its neighbors [95, 96, 92].

3.4 Hydrogen-bond and topology-based parameters

These parameters need hydrogen positions or a hydrogen-bond network. We use the geometric criterion of Luzar and Chandler: an O–O distance below 3.5 Å and an H–O$\cdots$O angle below 30° [97]. N_{HB} is the number of donated and accepted bonds. The four-body parameter $F_4 = \langle\cos 3\phi\rangle$ averages over neighbors the H–O$\cdots$O–H torsion formed with each molecule's outer hydrogens [98]; it is about -0.4 in ice, -0.04 in the liquid and 0.7 in clathrate hydrates [98, 99]. The selection of the outer hydrogen matters: choosing it per molecule from the opposite oxygen is the correct form, and we found and fixed an implementation that did otherwise. The Russo–Tanaka ζ is the distance to the nearest non-hydrogen-bonded oxygen minus that to the farthest hydrogen-bonded one [72]; it is positive in locally favored, tetrahedral structures. Ring statistics count the smallest rings of the hydrogen-bond network passing through each molecule [100, 101, 102]. Different ring definitions give different results [103], and in small periodic cells paths that wrap around the box must be rejected, or they are counted as rings.

3.5 Bond-orientational order

Steinhardt parameters [104] expand the directions to the neighbors in spherical harmonics,

$$q_{lm}(i) = \frac{1}{|N_b(i)|}\sum_{j\in N_b(i)} Y_{lm}(\hat{\mathbf{r}}_{ij}), \qquad q_l(i) = \sqrt{\frac{4\pi}{2l+1}\sum_{m=-l}^{l}|q_{lm}(i)|^2}, \tag{6}$$

and build the third-order invariant $w_l = \sum_{m_1+m_2+m_3=0}\begin{pmatrix} l & l & l \\ m_1 & m_2 & m_3\end{pmatrix} q_{lm_1}q_{lm_2}q_{lm_3}$ and its normalized form $\hat{w}_l = w_l/(\sum_m |q_{lm}|^2)^{3/2}$. Lechner and Dellago average the vectors, $\bar{q}_{lm}(i) = \sum_{j\in\{i\}\cup N_b(i)} q_{lm}(j)/(|N_b(i)|+1)$, and form $\bar{q}_l$ from them [79]; this picks up the orientation of the second shell. Odd l matter for water: $l = 3$ captures tetrahedral symmetry, and $\bar{q}_3$ and $\bar{q}_4$ distinguish Ih from Ic [105, 27]. The ten Wolde–Frenkel *crystalline connections* count neighbors j with normalized dot product $\mathbf{q}_6(i)\cdot\mathbf{q}_6(j) > 0.5$ [106]. For water, Reinhardt and co-workers

designed Steinhardt-based parameters able to drive ice growth [107], Haji-Akbari and Debenedetti labeled molecules with $q_6 > 0.5$ within 3.2 Å in their direct nucleation-rate calculation [10], and seeding studies tune a $\bar{q}_6$ threshold by a mislabeling criterion, which balances the fractions of mislabeled molecules in the bulk parent and nucleating phases [19, 108].

The CHILL algorithm [109] uses the correlation $c(i,j) = \mathbf{q}_3(i) \cdot \mathbf{q}_3(j)/(|\mathbf{q}_3(i)||\mathbf{q}_3(j)|)$ for each bond to the four nearest neighbors. Bonds with $c < -0.8$ are staggered and bonds with $-0.2 < c < -0.05$ eclipsed; four staggered bonds mean cubic ice and three staggered plus one eclipsed hexagonal ice, with further rules for interfacial ice. CHILL+ [110] widens the eclipsed window to $-0.35 < c < 0.25$, uses neighbors within 3.5 Å, and adds clathrate classes for molecules with four eclipsed bonds. It was designed to detect hexagonal ice up to its melting point.

3.6 Template and symmetry-based parameters

The centrosymmetry parameter sums $|\mathbf{r}_{ij} + \mathbf{r}_{ik}|^2$ over opposite pairs of the 12 nearest neighbors and vanishes for a centrosymmetric environment [111]. The neighbor-distance parameter D averages products of functions of the three distances in each neighbor triangle [92]. Polyhedral template matching (PTM) finds the template (here hexagonal or cubic diamond) with the lowest root-mean-square deviation after optimal rotation and scaling, and accepts it below an RMSD cutoff [112]; the related identify-diamond method uses common neighbor analysis of the second shell [113, 114]. The local-order metric of Martelli and co-workers measures the best overlap of a neighborhood with a reference lattice [115, 116].

3.7 Other recent parameters

Several recent parameters are not in our benchmark: an O–O–O–O dihedral parameter that separates Ih, Ic and high- and low-density liquid with Bayesian classification [117], the hydrogen-bond-network indicator Ψ [118], and an environment-similarity CV for oxygen-only ice growth [119]. Supervised screens over hundreds of candidate parameters and their combinations [91, 27, 28, 29, 120] and graph-network-designed parameters [121] have also produced task-specific sets.

3.8 Summary table

Table 2 lists every parameter in the benchmark with the settings used there. “Smooth form” marks a parameter defined through a switching function or Gaussian broadening. In our implementation neither marked parameter is a tested CV: both are truncated at the 5 Å search cutoff (where the switching function is small but not zero) and use hard neighbor membership.

Table 2: Local order parameters in the benchmark, with the settings used there. All O–O first-shell cutoffs are 3.5 Å. H: needs hydrogen positions.

Parameter	Group	Neighbors	H	Smooth form	Ref.
CN	radial	O within 3.5 Å			–
Smooth CN	radial	switching function, $r_0 = 3.5$ Å		✓	–
d_5	radial	fifth-nearest O			[81]
LSI	radial	O within 3.7 Å and next			[82]
Voronoi density, asphericity	radial	periodic Voronoi cell			[84, 85]
Entropy s, $\bar{s}$	radial	$g(r)$ to 5 Å; average to 5 Å		✓	[87, 88]
Q, $\bar{Q}$	angular	4 nearest O; average 3.5 Å			[90]
S_k	angular	4 nearest O			[89]
Bond angle B	angular	O within 3.5 Å			[93, 92]
Angular Fourier ($a = 3$)	angular	O within 3.5 Å			[94, 92]
Angle histogram H_1–H_3	angular	O within 3.5 Å			[92]
A_1–A_3	angular	O within 3.5 Å			[96, 92]
F_4	H-bond	O within 3.5 Å, outer H	✓		[98]
N_{HB}	H-bond	3.5 Å, 30°	✓		[97]
ζ	H-bond	H-bond partners; O to 5 Å	✓		[72]
Ring fractions (4–8), mean	H-bond	H-bond network, rings ≤ 10	✓		[100, 101]
q_3, q_4, q_6	bond-orient.	O within 3.5 Å			[104]
$\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$	bond-orient.	averaged over 3.5 Å			[79]
$\hat{w}_4$, $\hat{w}_6$	bond-orient.	O within 3.5 Å			[104]
Crystalline connections	bond-orient.	$\mathbf{q}_6$ dot product > 0.5			[106]
Centrosymmetry	template	12 nearest O			[111]
Neighbor distance D	template	O within 3.5 Å			[92]
CHILL / CHILL+	labeler	4 nearest / 3.5 Å, $\mathbf{q}_3$			[109, 110]
PTM	labeler	diamond templates, RMSD < 0.1			[112]

4 Descriptors and machine-learned approaches

4.1 High-dimensional descriptors

Machine-learning methods replace one scalar with a vector that describes the whole neighborhood. A useful descriptor is invariant to translation, rotation and permutation of identical atoms, changes smoothly with the coordinates, is complete enough to distinguish different environments, and is cheap to compute [122, 123]. Atom-centered symmetry functions sum radial and angular terms over neighbors [124]. The smooth overlap of atomic positions (SOAP) expands a Gaussian-smeared neighbor density in radial functions and spherical harmonics and forms the rotationally invariant power spectrum $p_{nn'l} = \sum_m c_{nlm} c_{n'lm}$ [94, 125]; Steinhardt q_l is closely related to it, retaining a single radial channel [122]. Many-body tensor representations encode distributions of distances and angles [126]. Libraries such as DScribe [127] make these descriptors easy to compute. For water, SOAP shows that the liquid contains local environments resembling a wide range of ice phases [76], that classical rigid models contain short-lived ice-like environments [77], and that SOAP and standard order parameters do not tell the same story about the liquid [128, 24].

4.2 Supervised classifiers

Supervised networks are trained on molecules of known phase. Geiger and Dellago trained a neural network on symmetry functions to identify liquid and several ice phases [129]. DeepIce

feeds the Cartesian coordinates of the nearest neighbors to a deep network and was applied to ice surfaces, where it outperformed Steinhardt, tetrahedral and template labels [130]. A PointNet-based network reached 99.1% accuracy over liquid, five ice phases and sI and sII hydrates using oxygen positions alone, about 99.6% with hydrogens, with most errors between sI and sII; with interfacial molecules included its accuracy fell to about 92% [25]. GCIceNet, a graph convolutional network on the hydrogen-bond graph with order parameters as node features, reported 99.8% test accuracy over nine TIP4P/Ice classes [26], and an edge-conditioned graph network 99.9% over nine phases, including the interface and water in nanotubes [16]. Score-based denoising can first remove thermal noise and then classify with simple parameters [131], and a variational autoencoder on SOAP features identifies ice phases without labels in its latent space [132].

These accuracies are not directly comparable. Each refers to its own classes, water model, temperature and train/test split; most test molecules come from the same trajectories as the training molecules; and accuracy on bulk phases says little about interfaces, which is where the classification matters most. The benchmark in section 5 shows that, with averaging, simple scalars already separate ambient ice from liquid with more than 99% balanced accuracy on held-out runs, so a learned model has to be judged on harder tasks: interfaces, high-pressure ice against dense liquid, and transfer to other water models.

4.3 Unsupervised approaches

Unsupervised methods look for structure in descriptor space without labels. Autoencoders and Gaussian mixtures on bond-order parameters recover crystalline and hidden local structures in colloidal systems [133, 134]; persistent-homology descriptors with clustering follow nucleation in metals [135, 136]; and a generalized convex hull with sketch-map projection organized nearly 75,000 hypothetical ice structures derived from zeolite networks and recovered the known phases among them [137]. Near the liquid–liquid critical point of a water model, local descriptors show no two-state structure while non-local ones reveal low- and high-density domains [138]. Unsupervised maps are descriptive. Turning a cluster into a phase label still requires reference structures, and the clusters depend on the descriptor and its cutoffs.

4.4 Data-driven collective variables

Enhanced sampling of nucleation needs a CV that is differentiable and that follows the slow structural change. Hand-built CVs for ice include Steinhardt-based parameters [139, 107], the averaged pair entropy [88], an environment-similarity kernel [119], and combinations tuned for crystallization [140, 47]. Data-driven CVs learn a low-dimensional combination of many descriptors, by discriminant analysis between basins (Deep-LDA) [141] or by the slowest modes of a biased trajectory (Deep-TICA) [142], and are typically combined with OPES [143]. The mlcolvar library exports such models to PLUMED [144]. Graph neural networks can learn nucleation CVs from Steinhardt features [145] or directly from coordinates [146], although neither has yet been tested on ice. In parallel, machine-learned potentials of first-principles quality [147, 148, 149, 150, 151] have brought nucleation studies to ab initio accuracy [152, 153, 12, 154]; they all still rely on a local order parameter to define the nucleus.

5 A unified benchmark

All parameters in table 2 were computed with one implementation (SolOrder, MIT license), one set of settings fixed before any comparison, and one analysis pipeline. Differences between parameters therefore reflect the parameters, not codes or cutoffs; the effect of the water model was measured separately. Before the benchmark, every parameter was checked against independent references: q_l against the Legendre addition formula (to 2×10^{-15}), direct geometric formulas for tetrahedral, radial and LSI parameters, MDAnalysis torsions for F_4, OVITO for CHILL+ (exact agreement on all GenIce phases tested), and serial against parallel execution on 366 frames from 44 nucleation trajectories.

5.1 Datasets

Ideal crystals. Eighteen structures were generated with GenIce2 [63, 64]: Ih, Ic, II, III, IV, V, VI, VII, VIII, IX, XI, XII, XIII, XIV and XVII, and the empty sI, sII and sH clathrate frameworks. Each proton-disordered phase was generated from five random seeds (89,100 molecular environments). Ice XVI was excluded because GenIce gives it the same oxygen positions as the empty sII framework. Every atom was displaced by Gaussian noise of 0.02 Å, about twice the coordinate precision of the files, so that two phases count as separable only at a stated resolution. Five pairs are proton order–disorder partners: Ih/XI, III/IX, V/XIII, VII/VIII and XII/XIV.

Seed test (run A). Bulk Ih, Ic and stacking-disordered ice, and spherical Ih, Ic and stacking-disordered seeds of 50, 100 and 200 molecules in liquid, were simulated with TIP4P/Ice [52] at 220 K at constant volume. Each of the twelve systems contributes ten thermal frames and the same frames after energy minimization (834,040 environments). Every molecule carries a label from the construction: seed membership, depth below the seed surface, and hexagonal or cubic stacking of its layer. The liquid reference consists of molecules more than 8 Å beyond the seed radius.

Replicas (run B) and independent preparations (runs C1–C5). Run B repeats bulk Ih, Ic, stacking-disordered ice and liquid (4096 molecules) at 220 K, and Ih and Ic at 250 K, from the same structures, with 60 frames 5 ps apart (1,643,520 environments). Runs C1–C5 start from new structures: Ih and Ic get new proton configurations, generated by ice-rule loop moves [155, 156] with the net dipole below 2% of full polarization; stacking-disordered ice gets a new random stacking sequence; each liquid is remelted at 300 K and re-equilibrated for 1 ns at 220 K (8,217,600 environments). Crystal oxygens stay within 0.35–0.41 Å of their minimized sites. The 220 K TIP4P/Ice liquid is slow: over the 300 ps of sampling the oxygen self-intermediate scattering function at $k = 2.25$ Å^{-1} decays only to 0.40–0.47, so its structural relaxation time exceeds 300 ps and is comparable to the 1 ns of re-equilibration. It does not age toward ice, however: the frame means of $\bar{q}_6$, $\bar{s}$, $\bar{Q}$ and smooth CN drift by at most 3% of their Ih–liquid separation over a run, and CHILL+ labels a constant 10% of molecules as ice or interfacial ice. The other liquids behave alike (drifts below 5% of the separation in every run), with faster relaxation for

TIP4P/2005 at 220 K ($1/e$ time 70–160 ps) and the liquids at 0.3–0.5 GPa (45–90 ps), and none at 3.0 GPa (still 0.78 after 300 ps, section 5.6).

Water models (run D). TIP4P/2005 [54] was run on the C1–C3 structures at 220 K (the same temperature) and at 200 K (about the same undercooling), together with a TIP4P/Ice control under the same protocol, so that moving a threshold from run C to the control measures the protocol effect and moving it from the control to TIP4P/2005 the model effect. Each system was equilibrated at 1 bar (anisotropic Monte Carlo barostat for crystals) and then sampled at constant volume (9,953,280 environments). The oxygen-only mW model [57], implemented in OpenMM [157] and checked against an independent energy calculation to 3×10^{-15}, was run on the C1–C3 oxygen positions at 220 K, close to the TIP4P/Ice undercooling (3,548,160 environments). Ih densities are 0.925 (TIP4P/2005), 0.912 (TIP4P/Ice) and 0.979 $\mathrm{g\,cm^{-3}}$ (mW).

High-pressure phases (run E). Ices II and III at 0.3 GPa, V at 0.5 GPa, VI at 1.0 GPa, and VII and VIII at 3.0 GPa were generated with GenIce2 (two seeds, every periodic height at least 24 Å), and liquid at each of these pressures (0.3, 0.5, 1.0 and 3.0 GPa) from the 4096-molecule liquid, remelted at 300 K and the target pressure, all with TIP4P/Ice at 220 K. Crystals were equilibrated for 200 ps at pressure (flexible barostat for the triclinic cells) and sampled as above (5,432,640 environments). All phases were retained: oxygens moved 0.25–0.45 Å over 300 ps, and the mean cosine between each molecule's dipole direction at the first and last frame stayed at 0.96, including the proton-ordered II and VIII. Densities were 1.18 (II), 1.15 (III), 1.26 (V), 1.35 (VI), 1.56 (VII) and 1.57 $\mathrm{g\,cm^{-3}}$ (VIII). Ice VII is metastable relative to VIII at 220 K and stays disordered only because rigid-water simulations do not order protons on this time scale.

Every thermal frame has a matching energy-minimized (quenched) frame, which removes thermal vibration while keeping the structure. Quenching uses 300 steps of FIRE [158] for every frame. Continuing to 10,000 steps on representative frames of each model and phase lowers the energy by at most 0.03 $\mathrm{kJ\,mol^{-1}}$ per molecule for the TIP4P models (residual net force per molecule 1–12 $\mathrm{kJ\,mol^{-1}\,nm^{-1}}$, against about 450 before quenching) and changes ice/liquid and Ih/Ic oracle scores by at most 0.015. The mW liquid converges more slowly (0.22 $\mathrm{kJ\,mol^{-1}}$ per molecule above the minimum after 300 steps), but its scores change by at most 0.026. Table 3 summarizes the coverage and the role of each dataset. The benchmark contains about 3.0×10^7 molecular environments, but that number measures computational coverage, not evidence: molecules in one frame, and frames of one trajectory, are strongly correlated. The inferential weight rests on the independent units, which are five preparations for the ambient phases at 220 K, three per water model and two per high-pressure phase.

Table 3: Coverage of the benchmark. Prep.: independent preparations (new proton configurations, stacking sequences or remelted liquids). Frames: thermal frames per run, each with a quenched copy. Everything not listed, including amorphous ices, thermal clathrates, ices IV, IX and XI–XVII at finite temperature, three-site water models and other temperatures and pressures, is outside the benchmark.

Dataset	Phases	Model	T, P	Prep.	Frames	Role
Ideal	18 GenIce structures	none	–	5 seeds[a]	1	structural reference
Run A	bulk Ih, Ic, Isd; seeds of 50–200 molecules in liquid	TIP4P/Ice	220 K, 1 bar	1	10, 1 ps	interface depth, labelers
Run B	Ih, Ic, Isd, liquid	TIP4P/Ice	220 K (Ih, Ic also 250 K), 1 bar	1[b]	60, 5 ps	held-out transfer
C1–C5	Ih, Ic, Isd, liquid	TIP4P/Ice	220 K (Ih, Ic also 250 K), 1 bar	5	60, 5 ps	main benchmark
Run D	Ih, Ic, Isd, liquid	TIP4P/2005; TIP4P/Ice NPT control; mW	220 K (TIP4P/2005 also 200 K[c]), 1 bar	3	60, 5 ps	model and protocol transfer
Run E	II, III, V, VI, VII, VIII; liquid at each pressure	TIP4P/Ice	220 K, 0.3–3.0 GPa	2[d]	60, 5 ps	high-pressure extension
Run F	Ih, Ic, Isd seeds of 50–1600 molecules in liquid; planar slabs	TIP4P/Ice	220 K, 1 bar	3	10, 2 ps	interface depth, nucleus-size bias
Run G	Ih, Ic, Isd seeds of 300, 600 molecules; prism and basal slabs	TIP4P/Ice	220 K, 1 bar	1[e]	10, 2 ps[f]	confirmation, facets, time
Sensitivity	Ih, Ic, liquid (C1–C3)	TIP4P/Ice	220 K, 1 bar	3	12, 25 ps	neighbor conventions
Nucleation	44 trajectories	TIP4P/Ice, mW	various	–	366 total	qualitative application

[a] For proton-disordered phases; ordered phases repeat. [b] Same starting structures as run A. [c] Ih, Ic and liquid. [d] Three for ice V and the 0.5 GPa liquid; the third is used only as the final test in the ice V case study. [e] A fourth preparation, built after the nucleus-size calibration had been frozen. [f] Long basal slabs: 34 frames, 2–500 ps.

5.2 Separability metric

For a scalar parameter and two populations A and B we use the two-sample Kolmogorov–Smirnov statistic $D_{AB} = \sup_x |F_A(x) - F_B(x)|$, where F_A and F_B are the empirical cumulative distributions over the molecules of each population. D needs no binning, which matters for the near-delta distributions of ideal crystals, and $(1 + D)/2$ is exactly the balanced accuracy of the best single threshold chosen on the same data. We report it as an *oracle score*, between 0.5 (no separation) and 1 (perfect separation). Because the threshold is optimized on the evaluated molecules, it is not a predictive accuracy. We therefore also report *held-out* scores: the balanced accuracy of a threshold and direction fitted on one run and applied to another, with missing values counted as errors. Molecules within one frame are correlated, so bootstrap intervals resample frames (blocks of five frames, 25 ps, in run B) rather than molecules. Such intervals describe variation within one trajectory and are therefore narrow; the spread between independent preparations is the better measure of uncertainty and is the one we quote. Labelers (CHILL, CHILL+, PTM) are scored against the construction labels.

5.3 Ideal crystals

Of the 153 pairs of ideal crystals, 150 are separated at an oracle score of at least 0.95 by at least one parameter (fig. 2). The three exceptions are Ih/XI (best 0.87, smooth CN), V/XIII (best 0.83, $\bar{q}_4$) and sI/sII (best 0.92). The first two are proton order–disorder partners whose oxygen frameworks barely differ, and no hydrogen-aware parameter (F_4, ζ, N_{HB}, ring fractions) separates them either. The other partner pairs are separable only because GenIce's ordered cells are slightly distorted: III/IX by $\bar{s}$ alone, VII/VIII by 17 parameters and XII/XIV by five. The most broadly useful parameters are $\bar{s}$ (93% of pairs at 0.95 or above), Voronoi asphericity (90%), Voronoi density and $\bar{Q}$ (89%). N_{HB} separates no pair, because every molecule in every ideal crystal has four hydrogen bonds; A_3 separates none either.

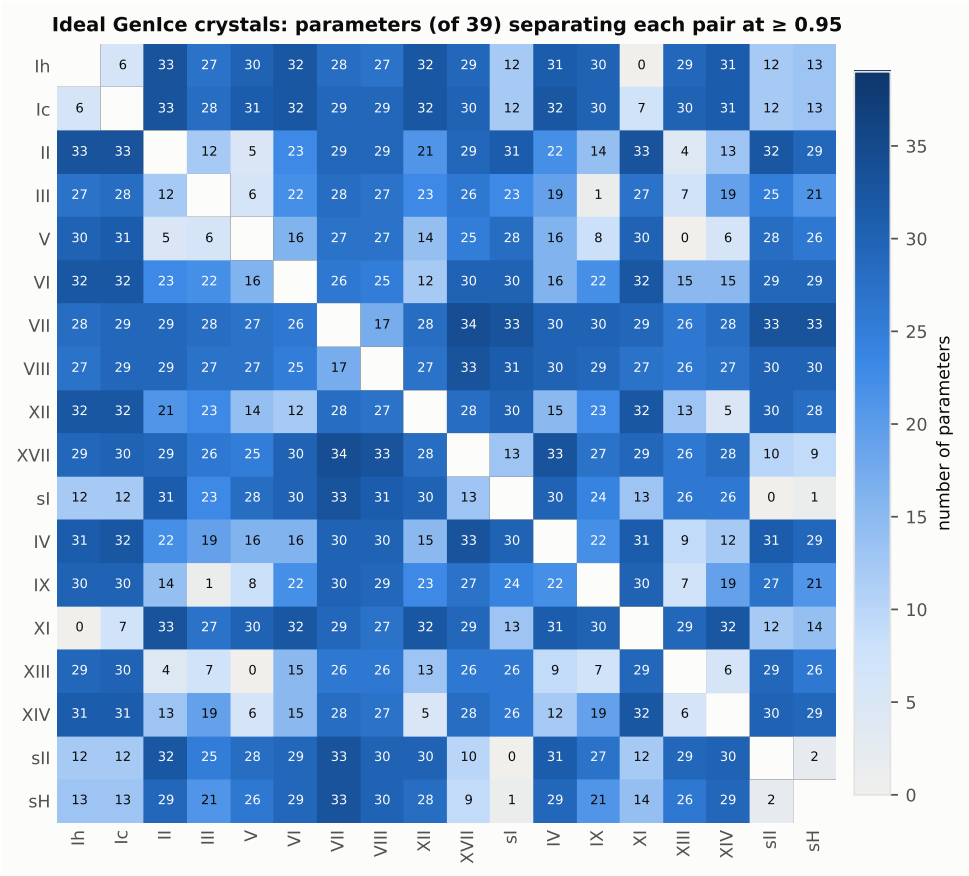


Figure 2: Number of parameters (of 39) that separate each pair of ideal crystals with an oracle score of at least 0.95.

5.4 Thermal motion: averaging is what survives

At 220 K the picture changes (fig. 3). For ice against liquid, the best parameters are locally averaged: $\bar{s}$ (1.00), $\bar{q}_6$ (0.99) and $\bar{Q}$ (0.98), compared with 0.85, 0.78 and 0.84 for their unaveraged forms. The six-ring fraction (0.99) is the best unaveraged parameter, since ring statistics already span several molecules. The cutoff coordination number (0.62) and N_{HB} (0.53) are nearly useless: supercooled liquid water is almost fully four-coordinated and hydrogen-bonded. Energy minimization restores most of the separation removed by thermal motion: d_5, LSI and ζ rise from 0.83–0.84 to 1.00.

Hexagonal and cubic ice are separated at 220 K only by $\bar{q}_3$ (1.00), $\bar{q}_4$ (1.00), F_4 (0.94), crystalline connections (0.93), $\bar{q}_6$ (0.78) and centrosymmetry (0.72); every other parameter is at 0.61 or below, including ring statistics, which cannot distinguish them. $\bar{s}$ separates Ih from Ic perfectly in ideal crystals but only at 0.61 at 220 K. At 250 K the order is unchanged ($\bar{q}_3$ 1.00, $\bar{q}_4$ 0.99, F_4 0.91, crystalline connections 0.88). These results agree with earlier findings that $\bar{q}_4$ separates Ih from Ic at the ice–water interface [105] and that low-l averaged parameters are the best pair in a supervised screen [27].

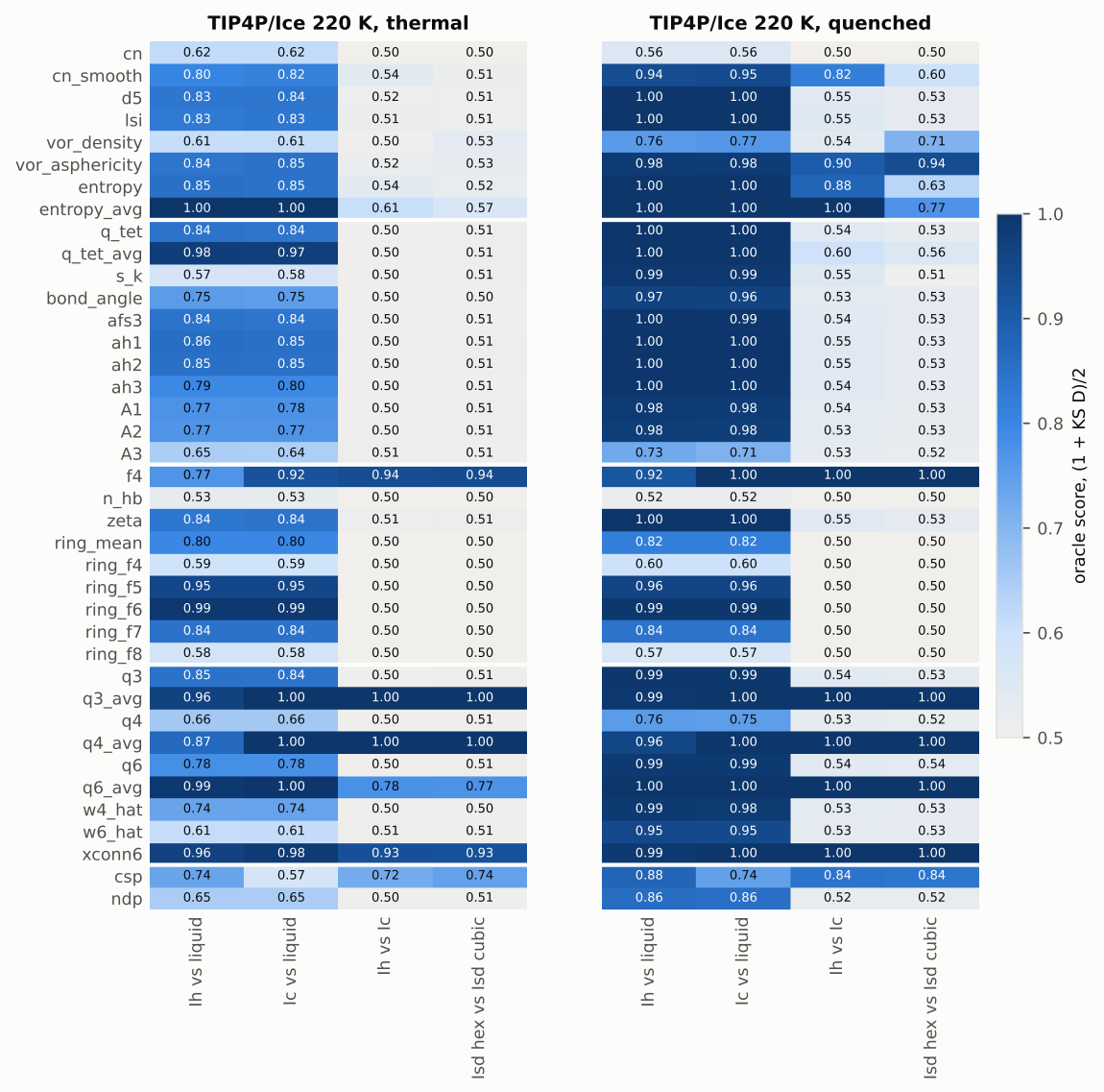


Figure 3: Oracle score $(1 + D)/2$ for ice versus liquid, Ih versus Ic, and hexagonal versus cubic layers in stacking-disordered ice, TIP4P/Ice at 220 K. Left: thermal frames; right: the same frames after energy minimization.

Many parameters carry nearly the same information in these data. Grouping parameters whose Spearman rank correlation, pooled over thermal Ih, Ic and liquid at 220 K, exceeds 0.8 in absolute value reduces the 39 columns to 22 groups (fig. 4). The largest joins Q, the bond angle, the angular Fourier term, the three angle-histogram amplitudes, q_3, q_6 and $\hat{w}_4$: in this mixture of ambient ice and liquid they rank molecules almost identically. Pooled correlations largely reflect

the ice–liquid contrast, so they show redundancy for that task, not equivalence in general. Other groups join $\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$ and crystalline connections; d_5, LSI, s and ζ; $\bar{Q}$ and $\bar{s}$; and the five- and six-ring fractions. Computed within single phases, most of this redundancy disappears. Of the 51 parameter pairs that share a pooled group, 80% stay above $|\rho| = 0.8$ within thermal Ih or Ic (where thermal distortions move them together), 27% within the liquid, and 0–4% within ices V, VI and VII or liquid at 1 GPa (median within-phase $|\rho|$ 0.14–0.21). The averaged parameters $\bar{q}_3$, $\bar{q}_4$ and $\bar{q}_6$, for example, correlate at 0.85–0.90 when pooled but only at 0.1–0.5 inside any single phase: they agree because they all respond to the ice–liquid contrast, not because they measure the same thing. A few pairs are redundant nearly everywhere, notably q_3 and the angular Fourier term with $a = 3$ ($|\rho| \geq 0.94$ in Ih, Ic, liquid and V, 0.88 in liquid at 1 GPa, but 0.4–0.6 in VI and VII).

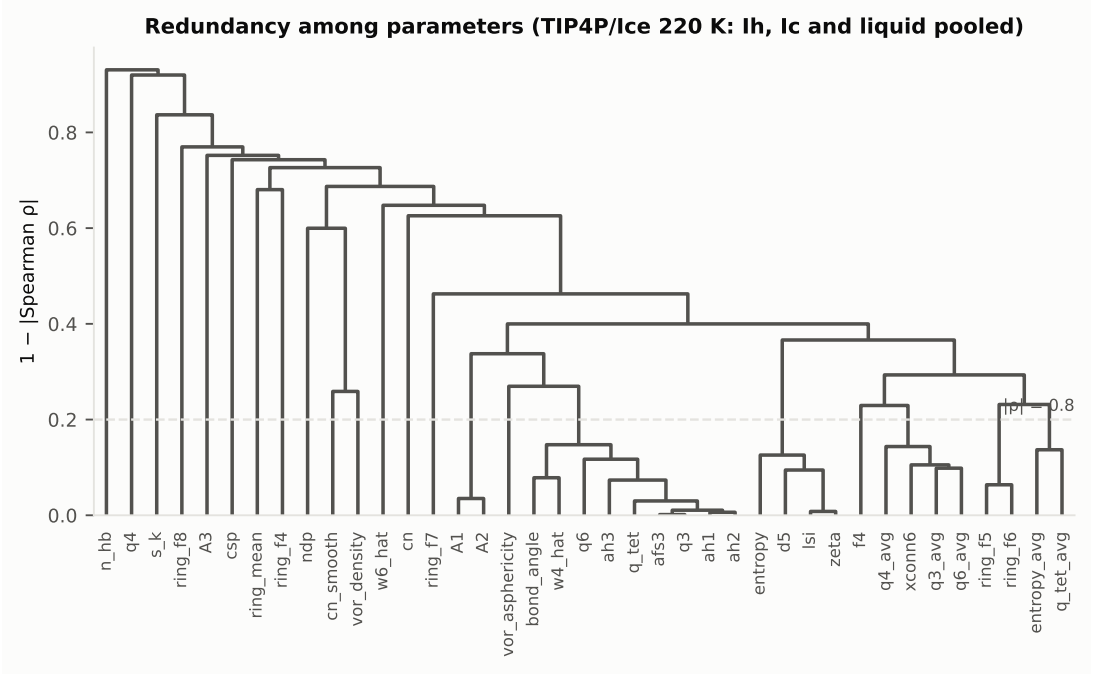


Figure 4: Hierarchical clustering of parameters by $1 - |\rho|$, with ρ the Spearman correlation over thermal Ih, Ic and liquid molecules at 220 K. The dashed line marks $|\rho| = 0.8$.

5.5 How far do thresholds transfer?

Between runs and preparations. A threshold and direction fitted on run A and applied to run B lose at most 0.008 in balanced accuracy relative to the oracle score on run B, for every parameter and population pair, thermal or quenched (fig. 5). With 10^4–10^5 molecules per population a single threshold barely overfits, and the best parameter for each task is the same under both scores. Across the five independent preparations the median standard deviation of the oracle score is 0.001; wherever the mean oracle score is at least 0.8, leaving one preparation out costs at most 0.001 and thresholds fitted on run A cost at most 0.003 (fig. 6). Hydrogen-aware parameters score within 0.009 of each other in the polarized run A crystals and the unpolarized preparations. Weak separations are another matter. In quenched stacking-disordered ice, smooth CN separates hexagonal from cubic layers at 0.60 in one preparation but 0.51–0.52 in the other four, and a threshold fitted on four preparations scores 0.45 on the fifth: the stacking sequence,

not the parameter, produced the separation. In these data, every separation that failed to reproduce across preparations was below about 0.65. We use that value as an empirical warning for these datasets, not as a general reliability threshold.

What matters more than the run is which populations are compared. With the bulk liquid of run B instead of the far liquid around the seeds of run A, 25 of 156 ice/liquid scores change by more than 0.01, all of them distance- or density-based (CN, smooth CN, d_5, Voronoi, bond angle, A_1, A_2, centrosymmetry, D), by up to 0.05.

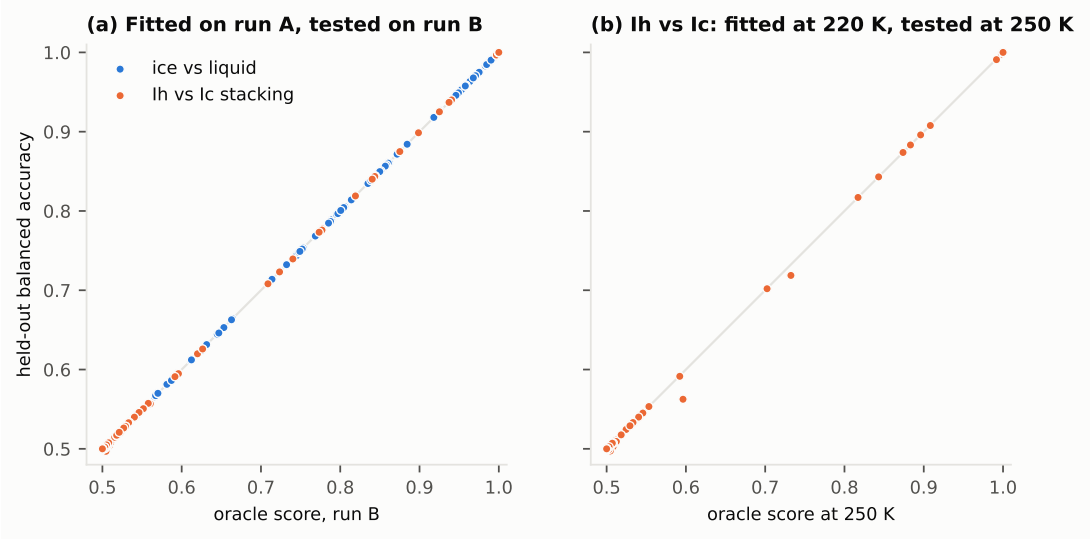


Figure 5: Held-out against oracle scores, one point per parameter and population pair (thermal and quenched). (a) Threshold fitted on run A, balanced accuracy on run B, at 220 K. (b) Ih versus Ic in run B: threshold fitted at 220 K, balanced accuracy at 250 K.

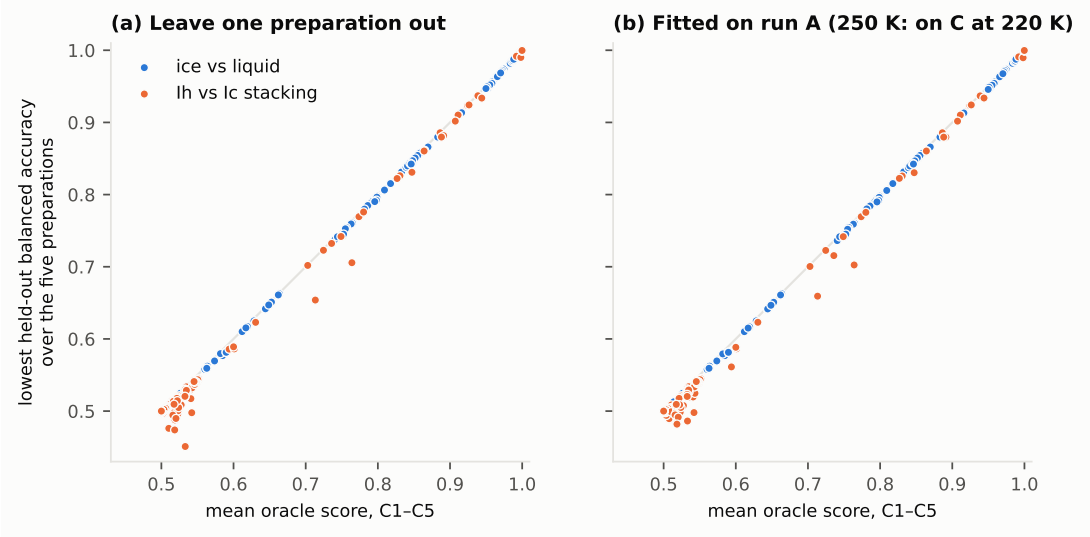


Figure 6: Five independent preparations (C1–C5). Vertical axis: lowest held-out balanced accuracy over the five preparations, for a threshold fitted (a) on the other four preparations and (b) on run A.

Between water models. Thermal frames transfer between the two TIP4P models (fig. 7a–c). From TIP4P/Ice to TIP4P/2005, oracle scores change by a median 0.004 (at most 0.025) at the same temperature or the same undercooling, and wherever the oracle score is at least 0.8

a threshold fitted on TIP4P/Ice loses at most 0.033 on the worst TIP4P/2005 preparation. Changing the protocol from the constructed constant-volume box to NPT costs at most 0.012. Quenched frames are the exception: their distributions are so narrow that the best threshold sits at a model-specific value. Smooth CN and Voronoi density keep oracle scores of 0.97 and 0.82 for ice versus liquid in either model, but thresholds moved from TIP4P/Ice score only 0.47–0.65 on TIP4P/2005.

The oxygen-only mW model is a harder test (fig. 7d). The leading parameters still transfer within 0.01: $\bar{s}$ (1.000), $\bar{q}_6$ (0.998), $\bar{Q}$ (0.991) and $\bar{q}_3$ (0.987) for ice versus liquid, $\bar{q}_3$ and $\bar{q}_4$ (at least 0.998) for Ih versus Ic. First-shell distance parameters, however, fail to transfer even in thermal frames, because the mW liquid packs more neighbors into the first shell (median smooth CN 4.76, against 4.39 for TIP4P/Ice). Smooth CN separates ice from mW liquid with an oracle score of 0.90 but scores 0.67 with the TIP4P/Ice threshold; A_1 drops from 0.80 to 0.72 and d_5 and LSI lose about 0.04.

Density artifacts in quenched frames. The narrowness of quenched distributions also turns small density differences into spurious structure. Under NPT, Ic comes out 0.35% denser than Ih in both TIP4P models, and that alone gives Voronoi density and D oracle scores of 0.74–0.90 for quenched Ih versus Ic, against 0.52–0.55 in a fixed box. In mW, with no proton disorder to blur the lattice, quenched Ih and Ic are separated almost perfectly (at least 0.998) by smooth CN, Voronoi density and asphericity, entropy and D. Neither is a stacking signal; both are lattice-parameter signals.

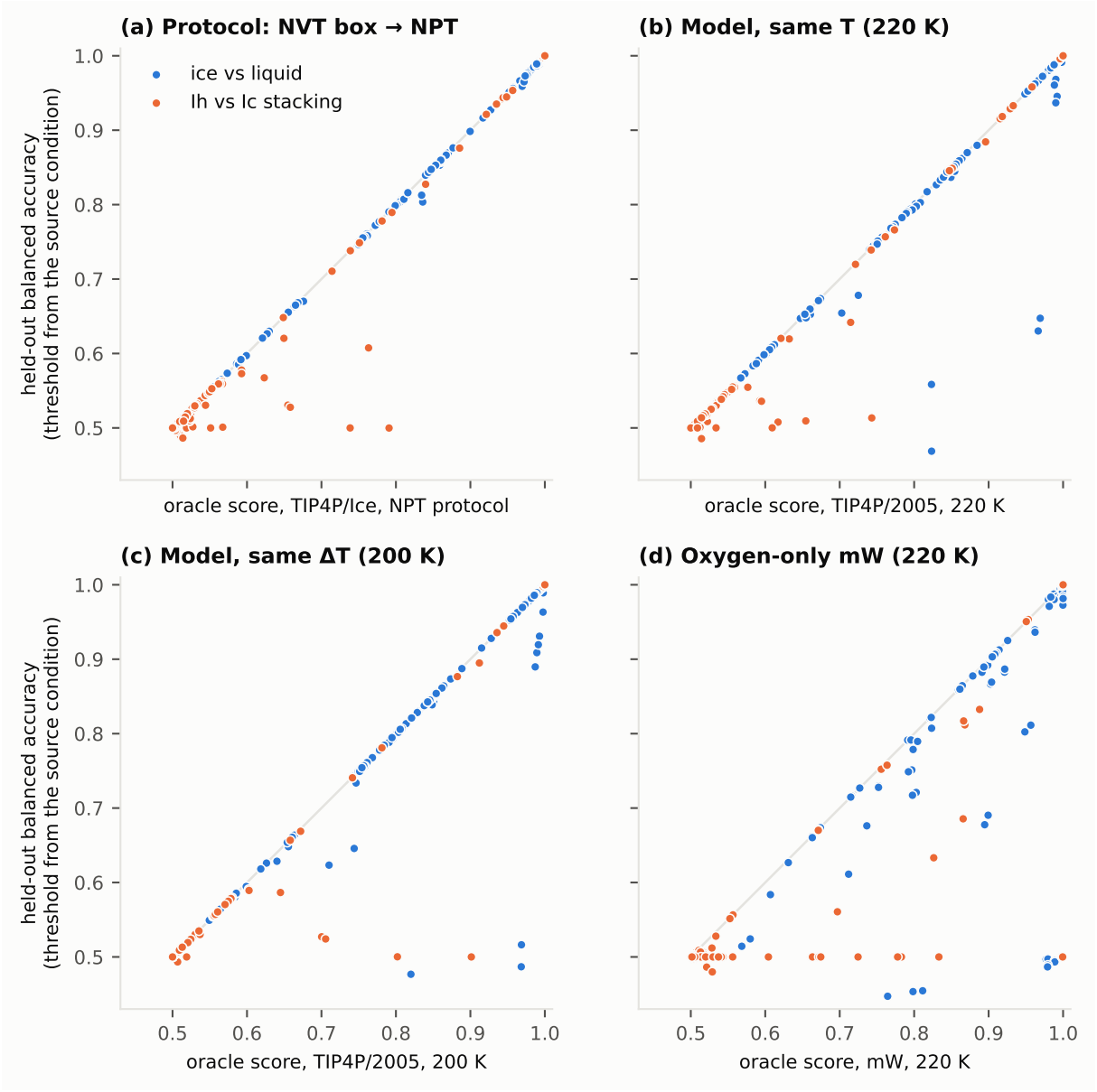


Figure 7: Moving thresholds between protocols and water models. Horizontal axis: oracle score on the target condition; vertical axis: held-out score of the threshold fitted on the source condition. (a) TIP4P/Ice, constant-volume box to NPT protocol. (b, c) TIP4P/Ice to TIP4P/2005 at 220 K and at 200 K. (d) TIP4P/Ice to mW at 220 K; hydrogen-dependent parameters are absent. The outliers far below the diagonal are distance- and density-based parameters, in quenched frames for TIP4P/2005 and also in thermal frames for mW.

5.6 High-pressure phases

Every pair of the eight thermal crystals (Ih, Ic, II, III, V, VI, VII, VIII) is still separated by some single parameter (fig. 8a): the best held-out score, taking the worse of the two preparations, is at least 0.98 for every pair except VII/VIII, where $\bar{q}_6$ reaches 0.92 (1.00 in quenched frames, through the angle histogram). Individual parameters suffer much more than this best case (fig. 8b). Of the 684 crystal pair and parameter combinations with an ideal oracle score of at least 0.95, 31% fall below 0.95 and 10% below 0.8 in thermal frames; quenching restores all but 3.5%. Losses are largest for $\hat{w}_4$ and $\hat{w}_6$ (median 0.23–0.26), D (0.14), and unaveraged q_6, F_4 and entropy

(0.06–0.08), and near zero for averaged and network parameters. $\bar{s}$ separates crystal pairs best on average (mean held-out 0.96), followed by smooth CN, Voronoi asphericity and $\bar{Q}$ (0.94).

Because the crystals were sampled at different pressures, a comparison of ice with liquid at another pressure mixes structure with state point. We therefore ran liquid at the pressure of every crystal and treat the pressure-matched pairs as the test of ice-versus-liquid identification; the other pairs measure discrimination among the sampled phase and state-point combinations. At 3.0 GPa and 220 K the liquid is nearly arrested (oxygens move 0.5 Å in 300 ps, compared with 2 Å at 0.3–0.5 GPa) and stays amorphous, with a density of 1.44 $\mathrm{g\,cm^{-3}}$ against 1.56 for ice VII, so it is closer to a compressed glass than to a liquid.

Matched against liquid at its own pressure, with the worse held-out score of the two preparations, ice VI (1.00, $\bar{s}$), VII and VIII (1.00, $\bar{s}$) are separated perfectly, II reaches 0.97 (six-ring fraction), III 0.95 ($\bar{s}$) and V only 0.84 ($\bar{q}_4$). Quenching raises V to 0.93. Ice V, with its 28-molecule cell and wide range of local environments, is the one phase that no single local parameter separates from liquid at its own pressure. No single parameter covers all matched pairs: the best worst case is $\bar{Q}$ at 0.69 (0.65 over all ice and liquid pairs). Liquids are hard to tell apart from each other at nearby pressures (0.64 between 0.3 and 0.5 GPa, 0.75 between 0.5 and 1.0 GPa, best with smooth CN). Ice-versus-liquid choices made at ambient pressure therefore do not carry over, and pairwise separation by one threshold does not imply that one classifier would sort all these phases at once.

Ice V as a case study. To find out what information ice V lacks, we ran a third, fresh preparation of ice V and of liquid at 0.5 GPa that no selection step saw. The best single parameter, chosen on the first half of preparation p1, is $\bar{q}_4$ (0.848 on p2, 0.847 on p3). Its errors are not confined to one kind of site: we characterized each ice V molecule by the number of oxygens within 3.5 Å in its time-averaged structure (5 to 9, because ice V folds non-bonded neighbors into the first shell) and by whether it lies on a four-membered ring, and the error rate is about 10% in every class. The reason is that the 0.5 GPa liquid is just as crowded (mean smooth CN 6.2, against 6.75 in ice V), so radial parameters alone (smooth CN 0.72) and averaged angular order alone (0.84) both overlap with it. Combining one of each helps. Among all 465 pairs of the 31 scalar parameters that are defined and vary in both phases, a two-feature logistic regression selected on p1 (fit on frames 0–29, validate on 30–59) picks smooth CN with $\bar{q}_4$; refitted on p1, it scores 0.923 on p2 and 0.922 on the untouched p3, and it cuts the error most for the most crowded molecules (from 11–12% to 3–6% for those with seven to nine close neighbors). The remaining 8% error may need medium-range information, which no parameter in the benchmark encodes. Thresholds are also fragile in quenched high-pressure frames: of the crystal pair and parameter combinations above, 62 quenched scores of at least 0.8 lose more than 0.02 between the two preparations (none in thermal frames), mostly for ices II and VIII. Quenched ordered crystals have nearly delta-shaped distributions (d_5 standard deviation 0.0002–0.003 Å), and the two preparations differ by their final box (0.4% in density for ice II), which moves the peaks by more than their width.

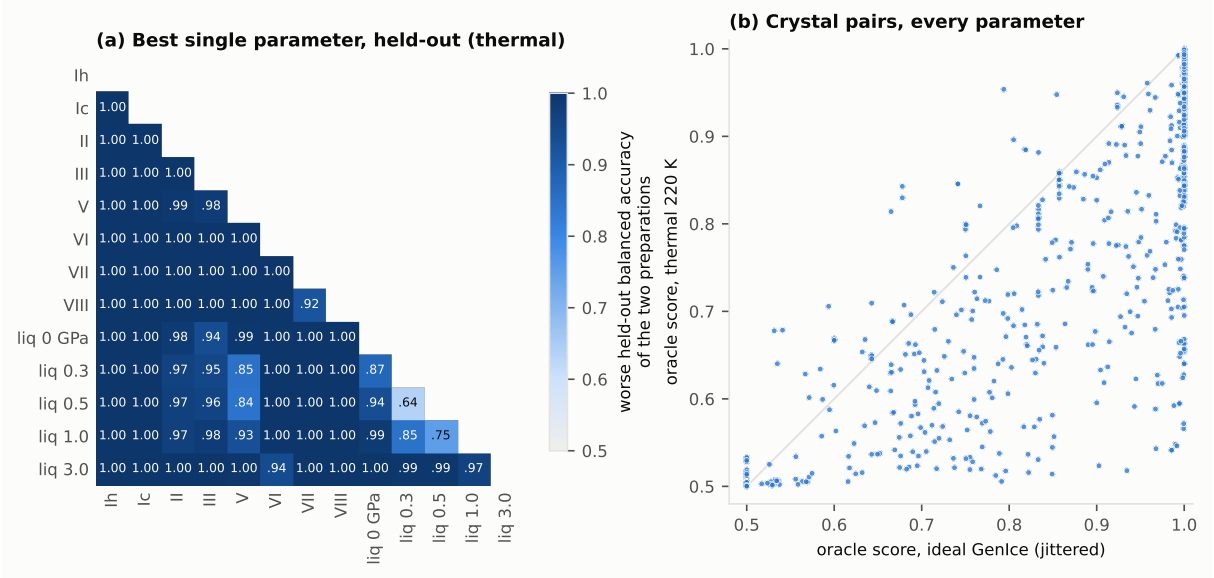


Figure 8: Thermal high-pressure phases, TIP4P/Ice at 220 K. (a) Best single parameter per pair, scored by the worse of the two held-out directions between preparations; liquid at the pressure of each crystal (0, 0.3, 0.5, 1.0 and 3.0 GPa; the 3.0 GPa liquid is nearly arrested). (b) Every parameter and crystal pair: oracle score of the ideal, jittered GenIce structures against the thermal oracle score.

5.7 Polytype recall against depth in constructed seeds

This section asks a narrower question than "does a labeler recognize ice": for molecules of a constructed Ih, Ic or stacking-disordered seed in TIP4P/Ice liquid at 220 K, how often does a labeler assign the bulk polytype of the molecule's layer, and how does that depend on the molecule's depth below the seed surface? A molecule that is not assigned the correct polytype can instead be assigned the wrong polytype, an interfacial-ice class (CHILL and CHILL+ only) or no ice class at all; these are different outcomes, and we report them separately. Each seed is one trajectory with ten frames 1 ps apart, so the intervals below describe variation within that trajectory.

In bulk ice, CHILL+ labels 99.7% of thermal Ih correctly, the original CHILL only 70.7%. The eclipsed-bond window explains the difference. In thermal bulk Ih every four-coordinated molecule keeps its three staggered bonds, but the correlation of its eclipsed bond falls inside CHILL's window $(-0.2, -0.05)$ for only 70.8% of molecules (5th–95th percentile -0.23 to 0.01), against 100% for the CHILL+ window $(-0.35, 0.25)$. After energy minimization both windows contain every eclipsed bond. This quantifies the improvement CHILL+ was designed for [110]. PTM with the default RMSD cutoff labels 97% of bulk ice correctly and never labels liquid as ice. False-positive rates in the far liquid are at most 1% for all three labelers.

In the seeds all three recover only part of the constructed crystal (fig. 9): CHILL+ finds 11% of the interior molecules of a 50-molecule Ih seed with the correct polytype, and 58% for a 200-molecule seed. Construction labels describe the starting structure, so we audited each frame before reading these numbers as errors. A seed molecule counts as intact when it lies within 1 Å of its constructed site (after removing the seed's drift) and keeps all its constructed bonds shorter than 3.5 Å. In the 100- and 200-molecule seeds 81–90% of seed molecules are intact (94–99% of interior ones); the 50-molecule Ih seed is the most deformed, 57% intact.

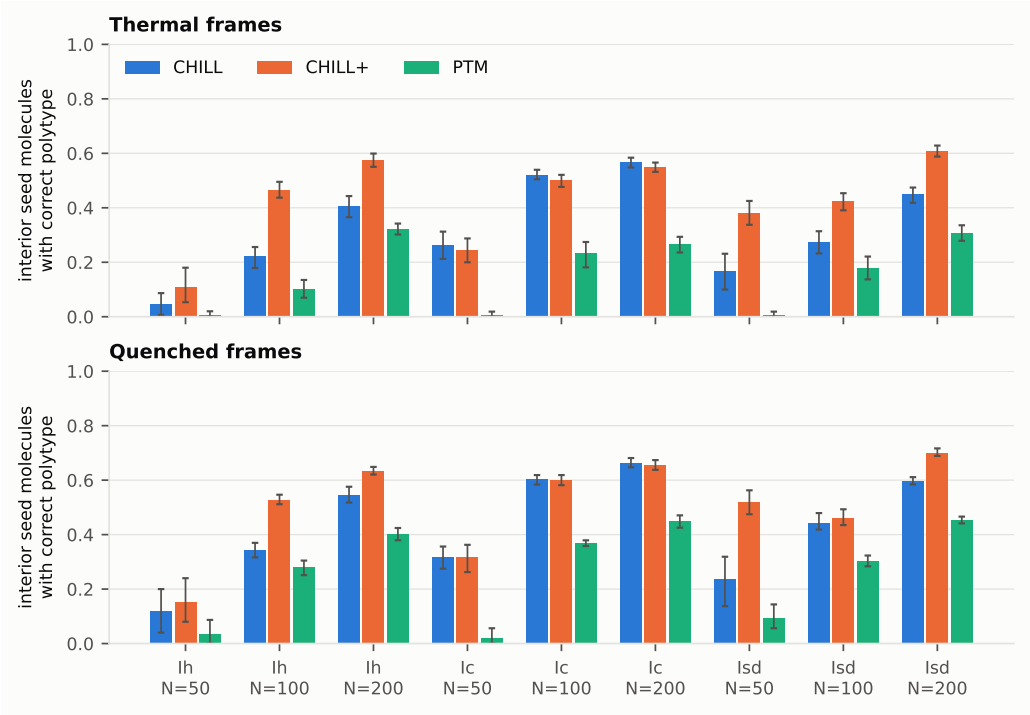


Figure 9: Fraction of interior seed molecules labeled with the correct polytype by CHILL, CHILL+ and PTM, for Ih, Ic and stacking-disordered (Isd) seeds of N molecules in TIP4P/Ice liquid at 220 K. Error bars: 95% bootstrap intervals over the ten frames of one trajectory per seed (1 ps apart); they do not include variation between independent seeds.

Polytype recall depends on depth below the constructed seed surface and hardly on seed size (fig. 10), but the three labelers behave differently. *CHILL+* assigns the correct polytype to at most 15% of molecules within 3 Å of the surface, 68–80% at 4–5 Å and at least 90% from 5–6 Å (Ic, stacking-disordered) or 6–7 Å (Ih), although about 90% of molecules are intact already at 2–4 Å. Most of the rest near the surface is not misassigned: at 3–4 Å it puts 44–47% of molecules in its interfacial-ice class, 15–25% in no ice class, and at most 7% in the wrong polytype. *CHILL* behaves like CHILL+ for Ic seeds, but for Ih and stacking-disordered seeds its recall stays at 0.5–0.9 even deep inside, the same eclipsed-window deficit as in bulk Ih. *PTM* has no interfacial class and never assigns the wrong polytype; it needs about 2 Å more depth than CHILL+, reaching 50–55% at 5–6 Å and 82–90% at 6–8 Å. A 50-molecule seed (radius about 7.3 Å) has only 3–4% of its molecules deeper than 5 Å, which explains most of its low score. Small-seed recall therefore measures how much crystalline support a labeler needs before it assigns a bulk polytype, not whether it detects ice; with its interfacial class, CHILL+ detects some form of ice at much smaller depths. The depths quoted hold for these constructed seeds in TIP4P/Ice at 220 K. This is consistent with the observation that labelers disagree mainly at interfaces [15], and it bears directly on nucleus sizes from seeding, which depend on where the labeler places the surface [19, 20, 21].

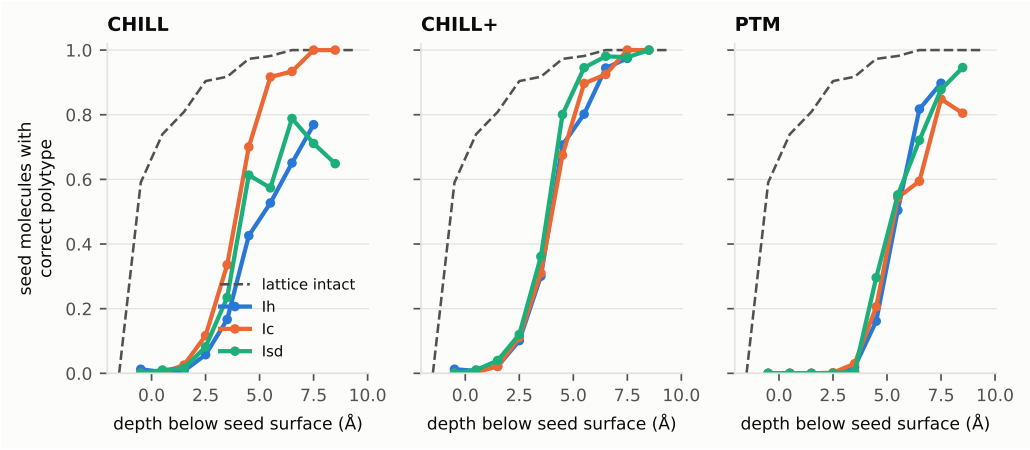


Figure 10: Polytype recall against depth: fraction of seed molecules labeled with the correct polytype against depth below the constructed seed surface, pooled over seed sizes, thermal frames at 220 K; molecules deeper than about 9 Å come only from the 200-molecule seeds. Dashed: fraction of molecules whose constructed lattice is intact. Bins with fewer than 30 molecules are omitted.

Replicated seeds, larger seeds and a planar control (run F). The seed test above rests on one trajectory per seed and seeds of at most 200 molecules. To turn it into a quantitative, testable result we built three independent preparations of Ih, Ic and stacking-disordered seeds of 50, 100, 200, 400, 800 and 1600 molecules (each cut around a random lattice site of a large crystal with its own proton configuration, randomly rotated, and inserted into its own 32,768-molecule liquid), and planar slabs of each polytype in contact with liquid, all in TIP4P/Ice at 220 K with the protocol of run A (restrained for 20 ps, then 10 frames 2 ps apart). The constructed crystals stay 73–94% intact, more for larger seeds, and the labeled fraction differs between preparations by at most 0.05 for seeds of 400 molecules or more.

Polytype recall at a given depth is not a property of the labeler alone: it depends on the curvature of the surface (fig. 11a, b). At the planar interface CHILL+ assigns the correct polytype to 26% of molecules 2.5–3 Å below the outermost oxygen plane, 87% at 3.5–4 Å and all molecules from 6 Å on, and PTM reaches 95% at 6–6.5 Å; a basal slab has molecules only at the depths of its layers (0, 2.76, 3.68, 6.44 Å, ...). In seeds the curves shift to larger depths the smaller the seed: a molecule at a given depth below a strongly curved surface has less crystal within the labeler's support. From about 400 molecules on they are close to the planar values, and summed over the seed's molecules slightly above them (see below). As a consequence, a recall-depth curve measured on seeds of 100 and 200 molecules does not predict other sizes. Summed over the constructed molecules of seeds it has not seen, it overestimates the correctly labeled count of 50-molecule seeds about twofold and underestimates seeds of 400–1600 molecules by 11–20% (CHILL, CHILL+) and 26–32% (PTM). A curvature-aware model, in which recall depends on the fraction of a probe sphere around the molecule that lies inside the crystal (probe radius 5–10.5 Å, fitted on the same small seeds), reduces these errors to 2–11% and 12–17%.

A planar calibration of finite-nucleus undercounting. The simplest description is also the most useful. We define the size of a nucleus by its equimolar radius, $R = (3N/4\pi n)^{1/3}$, with n the number density of the constructed lattice (0.0309 Å^{-3}; for an unknown nucleus, the density of the model's ice at the state point). From the planar slabs alone we measure, for each labeler

and observable, the bulk recall f (of molecules deeper than 8 Å) and the number of crystal molecules the labeler misses per unit area of one face, $\Gamma = \sum_i (1 - p_i/f)/A$, with p_i the labeled outcome of crystal molecule i. The dead layer is this count expressed as a thickness,

$$\delta = \Gamma / n, \tag{7}$$

which needs neither depth bins nor a reference plane. For the correctly assigned polytype $\delta = 3.4$ Å for CHILL+ and 5.0 Å for PTM (3.2–3.5 and 4.9–5.1 Å for the three polytypes, 3.2–3.7 and 4.7–5.3 Å for single slabs). A nucleus then has about

$$N_{\mathrm{lab}} \approx f\, N\, (1 - \delta/R)^3 \tag{8}$$

labeled molecules, and, inverting with R expressed through N, the size follows from the labeled count alone:

$$N \approx \frac{4\pi n}{3} \left[\left(\frac{3 N_{\mathrm{lab}}}{4\pi f n} \right)^{1/3} + \delta \right]^3 . \tag{9}$$

A depth integral, $\delta = \int [1 - p(z)/f]\, dz$ below the outermost oxygen plane, looks equivalent but is not well defined for a layered crystal: it depends on how the empty depth bins between layers are filled. For CHILL+ it gives 3.6, 3.0 or 2.4 Å with forward, linear or backward filling (PTM 5.5, 4.5 and 3.5 Å), and the pooled inverse error for CHILL+ becomes 8, 14 or 27% instead of the 5% obtained below; referred to the slab's equimolar surface, 0.9 Å further out, it overestimates the seeds by 27–35%. These alternatives are reported with the results; none is used.

No seed enters the calibration, so every seed tests it (fig. 11c, d). For seeds of 200–1600 molecules (36 systems per labeler: four sizes, three polytypes, three preparations), eq. (9) recovers the constructed size from the correctly assigned count with errors of 5% (largest error of the mean over preparations and polytypes), 5% (median over individual systems) and 10% (largest individual error) for CHILL+, and 10, 5 and 25% for PTM; eq. (8) predicts the counts themselves with 5, 6 and 16% (CHILL+). Applied to the observable nucleation studies actually report, the largest connected cluster of molecules labeled Ih or Ic (which can include liquid molecules), with f and δ calibrated for that observable on the slabs, the errors are 13, 2 and 19% for CHILL+ and 13, 4 and 28% for PTM. The pooled and largest errors come from the 200-molecule seeds, whose clusters are underestimated by 13% on average; from 400 molecules on, the mean error is at most 5% and no individual error exceeds 10% for either observable and labeler. The errors are systematic rather than random: correctly assigned counts of seeds of 400–1600 molecules are overestimated by 2–5%, and the mean signed error differs between preparations by less than one percentage point. The correction fails below 200 molecules, where R is less than three dead layers (underestimates of 9–39% on average). With the interfacial class, CHILL+ clusters give 6, 5 and 13%. CHILL is not suited to this calibration: its bulk recall depends on the polytype (0.70 for Ih, 0.86 for stacking-disordered ice, 1.00 for Ic), and one calibration for all polytypes leaves individual errors of up to 26%.

Confirmation on seeds prepared afterwards (run G). The dead-layer definition and the validated range were settled on run F after its seeds had been examined: the alternatives

above were scored on the same seeds. To test the calibration on data that played no part in any choice, we froze its code and values, recorded them with a pass criterion (errors within the range found for run F) and only then built and simulated a fourth preparation: Ih, Ic and stacking-disordered seeds of 300 and 600 molecules, sizes not used before, in a newly prepared liquid. All 24 predictions met the criterion. From the correctly assigned count, eq. (9) recovers the constructed size within 5% for CHILL+ and 7% for PTM. From the largest cluster it recovers it within 9% and 10%, with mean errors of −6% and −4%.

Facets and time. Seeds expose many facets, the calibration slabs only basal ones. Slabs with primary prism faces, built and simulated in run G with the same protocol, have a thinner dead layer, 3.05 Å for CHILL+ and 4.5 Å for PTM, about 10% below the basal values, which a fourth basal preparation reproduces (3.37 and 5.04 Å). Basal slabs followed for 500 ps after release keep the CHILL+ dead layer constant (slope 0.002 ± 0.005 Å per 100 ps); the PTM dead layer grows by 0.13 Å (3%) within the first 100 ps, which changes N by about 3% at 400 molecules. The intact fraction falls from 0.93 to 0.92 over the same time.

What the calibration assumes. Equations (8) and (9) are an approximation, not a geometric identity. Write the planar recall as $p(z) = f[1 - h(z)]$. If recall in a sphere of radius R and uniform density n depended on depth alone, the labeled count would be

$$N_{\mathrm{lab}} = 4\pi n f \int_0^R [1 - h(z)](R - z)^2 \, dz = fN \left(1 - \frac{3\delta}{R} + \frac{6\mu_1}{R^2} - \frac{3\mu_2}{R^3}\right), \qquad \mu_k = \int_0^\infty z^k h(z) \, dz, \tag{10}$$

with $\delta = \mu_0$ and h negligible beyond the sphere's centre. The leading, $1/R$, term is the number of missed molecules per unit area, which eq. (7) measures without reference to any plane; the higher moments depend on where depth is measured from and on the shape of the profile. Equation (8) is what eq. (10) becomes for a sharp missing shell, $h = 1$ for $z < \delta$ and 0 beyond ($\mu_1 = \delta^2/2$, $\mu_2 = \delta^3/3$), and it approximates these higher orders. Two effects beyond the leading order are seen in the data, and they act in opposite directions with size. Summing the basal recall at each seed molecule's depth underpredicts the labeled counts of seeds of 400–1600 molecules by 5–8% (CHILL+) and 8–12% (PTM): those seeds are labeled better than a basal surface at the same depth, consistent with the thinner dead layer of the prism faces they also expose. Below about 200 molecules the curvature effect dominates, and the same sum overpredicts 50-molecule seeds 2.7-fold (CHILL+). The correction is therefore empirical. It is exact only to leading order in δ/R, it rests on a basal calibration, and it has been validated only in the range given below.

The size it recovers is the constructed size. After release 6–14% of the constructed molecules of seeds of 200–1600 molecules (8–12% in run G) are no longer intact: they are more than 1 Å from their site or have lost a lattice bond. The slab dead layer includes the corresponding disorder of the interface. Against the intact count, eq. (9) overestimates by 12% (CHILL+, correctly assigned count) and 4% (CHILL+, largest cluster) on average, and by 10 and 8% for PTM. Neither count is a thermodynamic nucleus size.

The uncorrected bias is large. CHILL+, which labels 99.7% of bulk Ih correctly, assigns the right polytype to only 8, 19, 33, 47, 57 and 65% of the molecules of seeds of 50, 100, 200,

400, 800 and 1600 molecules, and PTM to 1–50%. Because δ differs between these labelers by 1.6 Å, the sizes they report for the same nucleus of a few hundred molecules differ by a factor of up to two. This is a spatial contribution to the dependence of nucleus sizes on the order parameter seen in seeding calculations [19, 20, 21]. Filion and co-workers absorbed the same dependence into a constant shift of the radius, fitted to umbrella-sampling barriers of hard spheres [30], and variational umbrella seeding makes barriers nearly independent of the order parameter [21]. The planar dead layer instead measures the shift directly, for each labeler and observable, without a barrier. The correction has been validated for correctly assigned and largest-cluster counts of constructed, approximately spherical Ih, Ic and stacking-disordered seeds of 200–1600 molecules in TIP4P/Ice at 220 K, observed 2–20 ps after release, with a basal calibration. It does not by itself give critical nucleus sizes, which also depend on how the nucleus is defined thermodynamically, and its transfer to faceted or growing nuclei, other temperatures and other models remains to be tested.

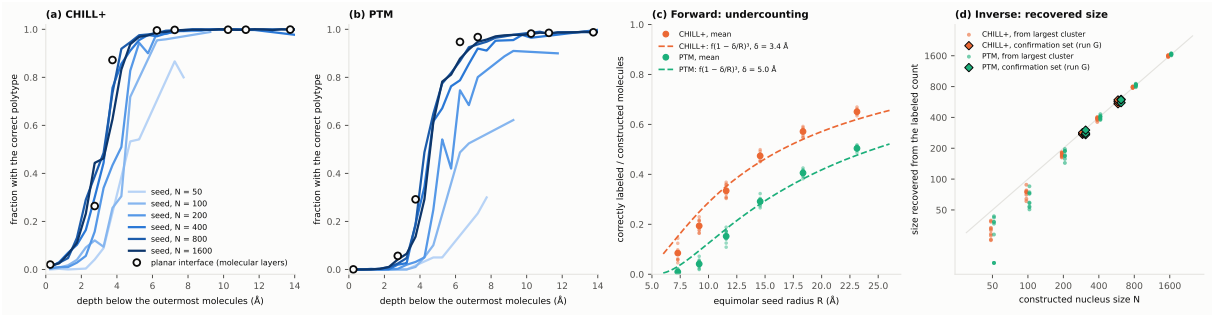


Figure 11: Planar calibration of finite-nucleus undercounting (run F; TIP4P/Ice, 220 K, Ih, Ic and stacking-disordered seeds, three preparations). (a, b) Fraction of constructed crystal molecules labeled with the correct polytype against depth below the outermost constructed molecules, pooled over polytypes, for seeds of 50–1600 molecules and for the planar basal interface (circles, at the depths of its molecular layers); bins with fewer than 30 molecules omitted. (c) Correctly labeled fraction against equimolar seed radius: small points, individual systems (one polytype, one preparation, mean over 10 frames); large points, mean per size; dashed, eq. (8) with f and $\delta = \Gamma/n$ from the planar interface only. (d) Size recovered by eq. (9) from the largest connected Ih/Ic cluster of each individual system against its constructed size; diamonds, the confirmation seeds of run G, prepared after the calibration had been frozen; diagonal, perfect recovery. Errors from the largest cluster per system: at most 19% (CHILL+) and 28% (PTM) for seeds of 200 molecules, at most 10% from 400 molecules on.

5.8 False nuclei in liquid references

A balanced per-molecule accuracy does not tell whether a labeling rule will report a nucleus where there is none. We therefore froze every rule before it saw the test data. Each scalar parameter got the threshold and direction that best separate thermal Ih from liquid in C1–C3; CHILL, CHILL+ and PTM kept their published settings. We applied the rules to liquids not used for fitting and measured, in every frame, the fraction of molecules labeled ice and the largest connected cluster of them (O–O edges below 3.5 Å; fig. 12).

In the held-out TIP4P/Ice liquid at 220 K (C4, C5, 120 frames of 4096 molecules), the recommended rules almost never form a false nucleus. $\bar{q}_6$ labels 0.7% of molecules as ice and its largest false cluster in any frame has 8 molecules; $\bar{s}$ labels 0.1% (largest cluster 7), CHILL+ 0.6% (4), and PTM none at all. Several rules with respectable per-molecule scores do much worse. Counting CHILL+'s interfacial class as ice labels 10% of the liquid and produces false clusters

of 14–17 molecules on average and up to 30, of which 30–45% are again part of a cluster 5 ps later. Smooth CN (27% false positives) forms clusters of up to 700 molecules, which connect to their own periodic image in 5–12% of frames, and d_5 and $\bar{q}_4$ (20%) form clusters of up to 75 and 98 molecules, comparable to critical nuclei at moderate supercooling. These rules fail because their best balanced threshold accepts a fifth of the liquid. With a threshold that labels 1% of the fitting liquid as ice instead, their largest false clusters shrink to at most 6 molecules, but they then label only 0.5% (smooth CN), 15% (d_5) and 3% ($\bar{q}_4$) of held-out thermal Ih as ice, whereas $\bar{q}_6$ and $\bar{s}$ keep 99% at that threshold. In this regime the largest false cluster does not grow with system size. In four independent liquids of 32,768 molecules (one configuration each), $\bar{q}_6$, $\bar{s}$, CHILL+ and PTM give at most 4, 2, 4 and 0 molecules, and smooth CN 144–287 molecules, comparable to the typical largest cluster of one 4096-molecule frame (240–260). With the TIP4P/Ice thresholds unchanged, TIP4P/2005 liquids at 220 and 200 K and the mW liquid behave like the TIP4P/Ice liquid for the recommended rules: their largest false clusters have at most 15 molecules ($\bar{s}$, TIP4P/2005 at 200 K) and at most 11 for the others. Frozen thresholds fail in dense liquids instead. Crystalline connections, whose count of correlated $\mathbf{q}_6$ bonds rises with density, report false clusters of up to 18–47 molecules at 0.3–1.0 GPa. In the nearly arrested liquid at 3.0 GPa they report clusters of about 580 molecules on average, which connect to their own periodic image in up to 78% of frames, and there $\bar{s}$ also finds false clusters of up to 77 molecules.

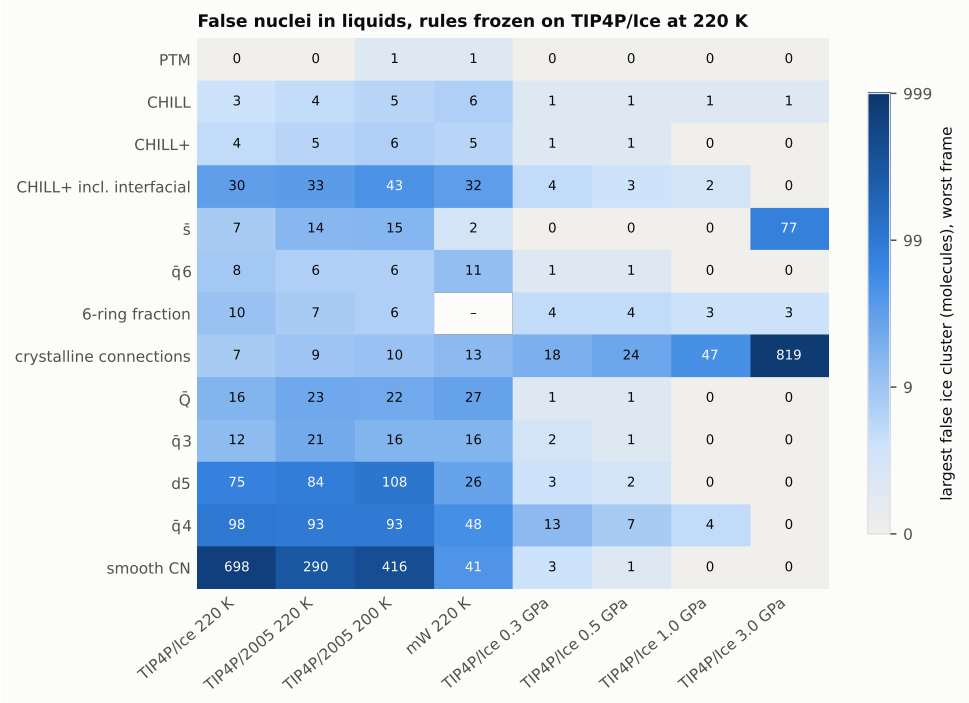


Figure 12: Largest connected cluster of molecules labeled ice in liquids that were not used for fitting, with rules frozen on TIP4P/Ice at 220 K (thresholds from C1–C3 thermal Ih versus liquid; labelers at their published settings). Largest cluster in any frame of any preparation (60 frames each; TIP4P/Ice 220 K: C4 and C5).

5.9 Application: nuclei along nucleation paths

As a qualitative check outside the benchmark data, we computed all parameters on 366 frames from 44 nucleation trajectories (biased path sampling, metadynamics, shooting and brute-force runs, TIP4P/Ice and mW). In the 34 frames of four fixed-box TIP4P/Ice path-sampling trajectories whose largest connected CHILL+ ice core holds at least 20 molecules, we compared each parameter with membership of that core. The median ROC area under the curve is highest for $\bar{q}_6$ (0.985), followed by $\bar{Q}$ and $\bar{q}_3$ (0.97), $\bar{s}$ (0.96, lower inside the core), crystalline connections (0.95) and $\bar{q}_4$ (0.93); N_{HB} is near 0.5. The ranking matches the bulk benchmark. Because CHILL+ is itself built from $\mathbf{q}_3$ correlations, this is agreement with a reference labeler, not independent accuracy, and the biased paths give no rates.

5.10 Beyond ice: confined water

The same parameters are often used to single out functionally important water in biomolecular simulations. As a transfer test outside the benchmark's design, the supplementary material applies them unchanged, computed from water oxygens and hydrogens only, to ligand-unbinding paths of the host–guest complex OAMe–G2 (TIP3P) and of T4 lysozyme L99A with benzene (SPC/E), both at 300 K, with geometric definitions of the waters of interest that no order parameter enters. Held out across paths or replicas, Voronoi density and smooth CN separate isolated cavity waters from bulk and protein-surface water at 0.91–0.99, but buried waters in connected internal clusters only at up to 0.80, and no parameter adds to position in predicting which waters enter a cavity. The parameters respond to the missing water neighbors of confined water, not to a distinct local structure; the large $\bar{q}_6$ of isolated waters is an artifact of having one or two neighbors.

5.11 Sensitivity to the neighbor convention

All results above use the registry defaults. To see how much they depend on that choice, we varied one factor at a time on preparations C1–C3 at 220 K (bulk Ih, bulk Ic and liquid, 12 thermal frames 25 ps apart and their quenched counterparts; 571,392 environments): the first-shell cutoff of every cutoff-based parameter and of CHILL and CHILL+ from 3.2 to 3.8 Å, the LSI shell radius over the same range, the averaging radius of $\bar{q}_l$ and $\bar{Q}$ from 3.2 to 5.0 Å with q_{lm} and Q kept at 3.5 Å, and the integration radius r_m of the pair entropy, which in our implementation is also its averaging radius, from 4.0 to 6.0 Å. Rank-based parameters (d_5, Q, S_k, centrosymmetry) and Voronoi parameters have no cutoff. Besides the oracle score we fitted a threshold on C1 at the default setting and applied it to C2 and C3 computed with the varied setting, which is what happens when a published threshold is reused with a different convention (fig. 13).

First-shell cutoff. The leading parameters barely notice it. For thermal Ih versus liquid, $\bar{q}_6$ stays at 0.987–0.992 and the six-ring fraction at 0.989–0.992; for Ih versus Ic, $\bar{q}_3$ stays at 0.983–1.000, $\bar{q}_4$ at 0.993–0.997, F_4 at 0.930–0.939 and crystalline connections at 0.906–0.925, all lowest at 3.8 Å. Their thresholds fitted at 3.5 Å lose at most 0.025 at any other cutoff. The ranking of the 30 scanned columns is stable (Spearman correlation with the default ranking

at least 0.94 for ice versus liquid and 0.99 for Ih versus Ic), and the three best parameters for Ih versus Ic are the same at every cutoff. Unaveraged first-shell parameters are another story. For thermal Ih versus liquid, 15 of the 30 columns change by more than 0.02, all of them rising with the cutoff: CN from 0.53 to 0.81, smooth CN from 0.63 to 0.87, D from 0.57 to 0.76, the bond-angle parameter from 0.73 to 0.85 and A_1 from 0.76 to 0.87. A larger cutoff reaches the interstitial molecules that distinguish the liquid, so none of these parameters has a natural cutoff, and published values computed with different cutoffs cannot be compared. Smooth CN is the extreme case: its threshold fitted with $r_0 = 3.5$ Å scores only 0.50–0.69 at any other r_0. Hydrogen-bond parameters (N_{HB}, ζ, ring fractions) and the LSI shell radius hardly matter (LSI 0.832–0.834), because the angular part of the hydrogen-bond criterion, not the distance, decides which pairs count.

Averaging radius. Averaging further than the first shell improves the even-l parameters: for thermal Ih versus liquid, $\bar{Q}$ rises from 0.976 to 1.000 and $\bar{q}_4$ from 0.870 to 0.923 at 5 Å, and for Ih versus Ic $\bar{q}_6$ rises from 0.776 to 0.962. $\bar{q}_3$ does the opposite. Its perfect Ih/Ic separation exists only with first-shell averaging and falls to 0.894, 0.647 and 0.804 at averaging radii of 4.0, 4.5 and 5.0 Å. The $\mathbf{q}_3$ vectors of neighboring molecules in a tetrahedral network are anti-correlated (the staggered bonds of CHILL), so the average depends on how many molecules of each sublattice it contains, and the second shell adds them in a different proportion. Thresholds of $\bar{q}_3$, and of $\bar{q}_4$ for ice versus liquid, do not survive a change of averaging radius at all (0.50).

Entropy radius. $\bar{s}$ separates thermal ice from liquid at 0.95 with $r_m = 4$ Å and 1.00 from 5 Å on; it never separates thermal Ih from Ic (at most 0.63). In quenched frames it separates Ih from Ic perfectly at the default $r_m = 5.0$ Å but only at 0.53–0.77 at any other radius: the integration limit happens to fall where the sharp quenched radial distributions of the two polytypes differ. The perfect quenched separation in fig. 3 is an accident of the cutoff, not a property of the parameter. Thresholds of s and $\bar{s}$ are tied to r_m (0.50 when it changes).

Labelers. CHILL+ labels 99.7% of thermal bulk Ih correctly with cutoffs of 3.3–3.5 Å, 98.6% at 3.7 Å and 96.1% at 3.8 Å, because it labels only molecules with exactly four neighbors inside the cutoff. CHILL uses the four nearest neighbors and is insensitive to the cutoff (70.7–71.0% for Ih, at least 99.3% for Ic). Both keep false positives in the liquid below 1% at every cutoff.

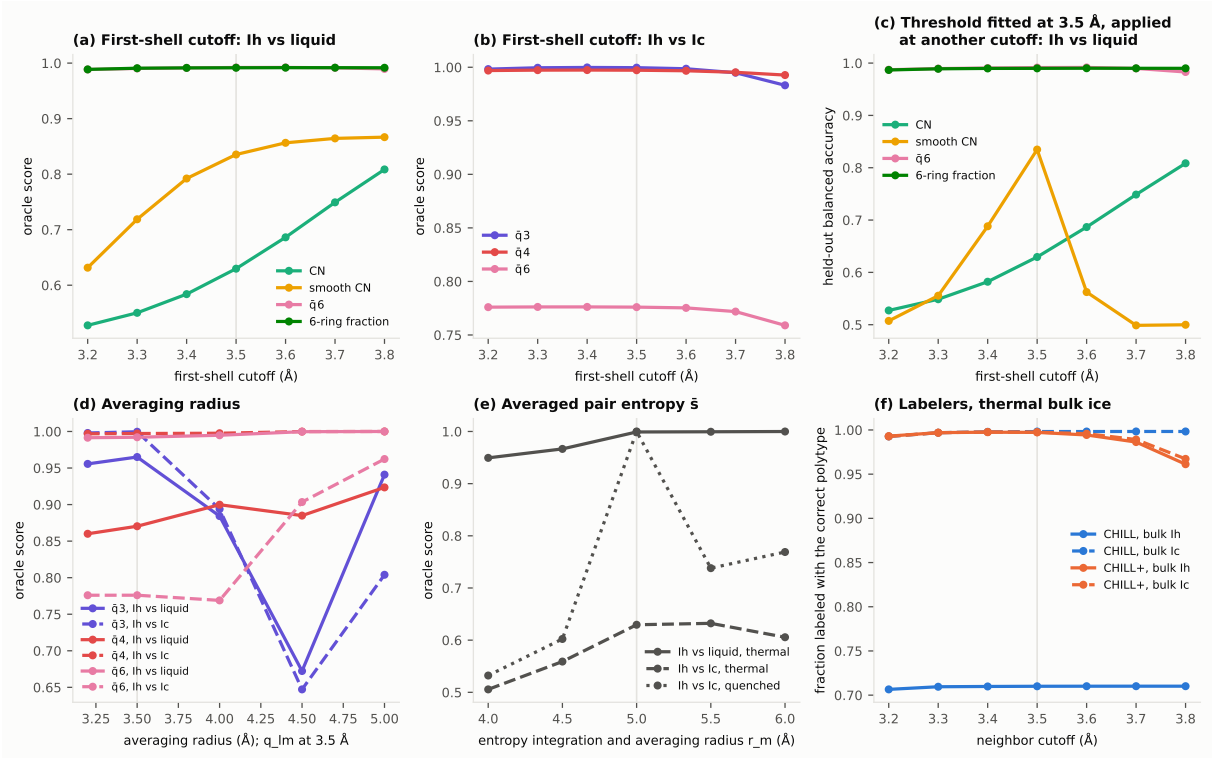


Figure 13: Sensitivity to the neighbor convention, run C1–C3 at 220 K. (a, b) Oracle score against the first-shell cutoff of every cutoff-based parameter ($\bar{q}_6$ and the six-ring fraction overlap near 0.99 in (a)). (c) Held-out score of a threshold fitted on C1 at 3.5 Å and applied to C2 and C3 computed with another cutoff (worse of the two). (d) Averaging radius of $\bar{q}_l$, with q_{lm} at 3.5 Å. (e) Integration and averaging radius of $\bar{s}$. (f) Fraction of thermal bulk ice labeled with the correct polytype by CHILL and CHILL+. Thin vertical lines mark the defaults.

5.12 Computational cost

We timed every registry entry on periodic tilings of one thermal frame of C1 (Ih: 1920 to 61,440 molecules; liquid: 4096 to 65,536) on one Intel Core i7-12700F, starting each measurement from a fresh analysis snapshot so that the neighbor search is included, as in the benchmark itself (fig. 14). Every entry scales linearly with system size (fitted exponents 0.99–1.04; PTM is sublinear because of a fixed setup cost). On one thread the whole registry costs about 125 μs per molecule, 8.2 s for a 65,536-molecule frame. The cheapest entries are CN (1.7 μs per molecule), PTM (2.7 μs), CHILL and CHILL+ (5 μs) and A_1–A_3 (5 μs); most others cost 6–13 μs, largely the neighbor search to 5 Å. The most expensive are ring statistics (29 μs), the angle histogram (20 μs), Voronoi (19 μs) and the pair entropy (17 μs). Cost is therefore not a reason to prefer a weaker parameter: the recommended $\bar{s}$, $\bar{q}_l$ and $\bar{Q}$ cost 8–17 μs per molecule, a negligible fraction of the simulation that produced the frame.

Threads help less than one might expect. With eight threads, ring statistics run 3.1 times faster, crystalline connections 2.5 times, Steinhardt parameters 2.4 times and A_1–A_3 2.1 times, but the full registry gains only a factor of 1.6 at 65,536 molecules, and below about 8,000 molecules it is slower than on one thread, because the neighbor search and several kernels run serially. For trajectories, parallelism over frames is the more effective choice; the benchmark itself ran ten frames at a time in separate processes.

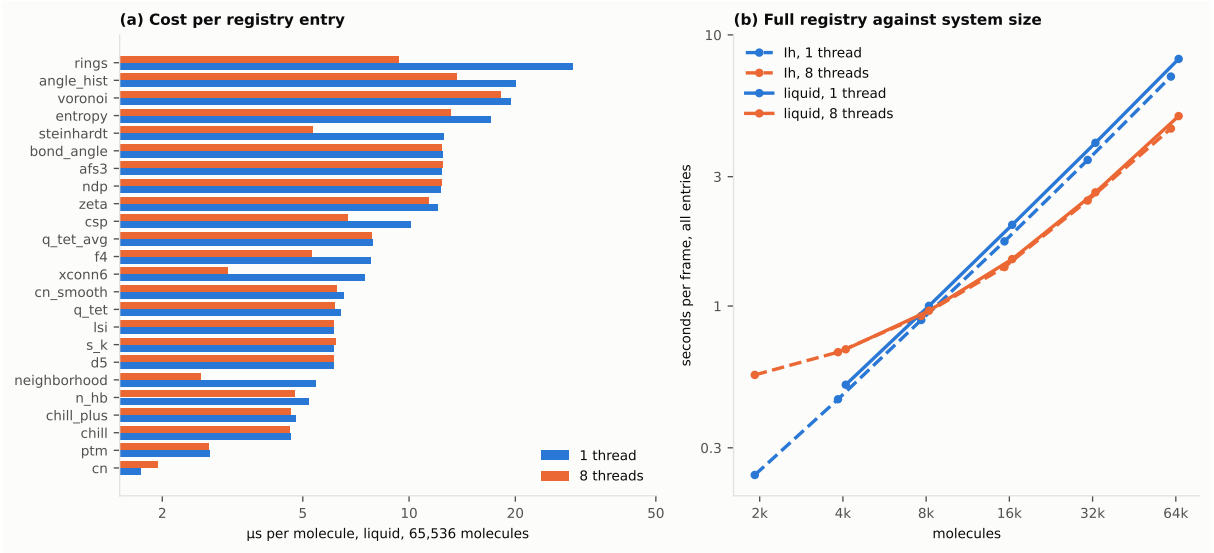


Figure 14: Computational cost on an Intel Core i7-12700F, median of three warm repeats, each from a fresh snapshot (neighbor search included). (a) Time per molecule for each registry entry, liquid with 65,536 molecules, one and eight threads. (b) Time per frame for all entries against system size; Ih solid, liquid dashed.

5.13 What the benchmark does not cover

Table 3 defines the boundaries of the benchmark. Within them, several questions remain open. The sensitivity scan covers one temperature and the ambient phases only, and varies one factor at a time. Only one temperature is available for most phases, and one pressure for each high-pressure phase. Ices IV, IX and XI–XVII and the empty clathrate frameworks were studied only as ideal structures; amorphous ices were not studied at all, and no three-site water model was included in the ice benchmark. Pairwise separation by one threshold does not establish that a single multiclass classifier would work for all phases at once. The 220 K TIP4P/Ice liquid relaxes on a time comparable to its equilibration (it shows no aging in the order parameters, but it is not far beyond one relaxation time), and the 3.0 GPa liquid is nearly arrested. The interface-depth results hold for TIP4P/Ice at 220 K, up to 500 ps after release and for basal and primary prism faces; at other temperatures, on other facets, or for growing and melting interfaces, the dead layer may differ. The ice V analysis characterized molecules by their local coordination rather than by crystallographic site, because proton disorder distorts every site differently. Ideal-crystal results for proton-ordered phases depend on the GenIce cell geometries. Finally, scoring per molecule does not test a parameter as a CV; that requires continuity at neighbor exchanges, correct gradients and, ideally, committor analysis.

6 Practical guide

Table 4 condenses the benchmark into recommendations by task. Seven rules of thumb follow from it.

1. *Average.* At finite temperature, averaging over the first shell is worth more than any other design choice. Averaged entropy, $\bar{q}_6$ and $\bar{Q}$ separate ambient ice from liquid at 0.98–1.00;

their unaveraged forms at 0.78–0.85.

2. *Use bond orientations for stacking.* Only averaged low-l Steinhardt parameters, F_4 and crystalline connections see the difference between Ih and Ic in thermal frames; rings and radial parameters do not.

3. *Do not move thresholds of distance- or density-based parameters* between water models, boxes, protocols or cutoffs without recalibrating them, and be suspicious of separations found only in quenched frames.

4. *Keep the averaging radius with the threshold.* The first-shell cutoff barely affects averaged and bond-orientational parameters, but the averaging radius and the entropy radius change their values and, for $\bar{q}_3$, their meaning. Report both and recalibrate when either changes.

5. *Recalibrate for high pressure.* No single parameter separates all high-pressure ices from liquid at their own pressure; choose by pair, and compare against liquid at the same pressure. Where one parameter is not enough, as for ice V, pair a radial with an averaged angular parameter.

6. *Check for false nuclei, not only accuracy.* Apply the frozen rule to a liquid at the state point of interest and report the largest false cluster. Rules with good per-molecule scores (smooth CN, d_5, $\bar{q}_4$, CHILL+ with its interfacial class) form false clusters of tens of molecules or more in supercooled liquid, and density-sensitive rules such as crystalline connections do so in compressed liquid.

7. *Correct nucleus sizes for the labeler's dead layer.* Measure, once on a planar interface and for the observable you report (correct polytype or largest cluster), the bulk recall f and the number of crystal molecules the labeler misses per unit area. Convert that number to a dead layer $\delta = \Gamma/n$ (eq. (7)) and recover the size with eq. (9), R being the equimolar radius. This was validated for spherical seeds of 200–1600 molecules in TIP4P/Ice at 220 K and confirmed on seeds prepared afterwards. For single seeds of 200 molecules the errors reach 10% (correct-polytype count) and 19% (largest cluster) with CHILL+, and 25 and 28% with PTM. From 400 molecules on they stay within 10% for both labelers. The size it recovers is the constructed size, which exceeds the intact crystal by about 10%. Check it before relying on it for smaller, faceted or growing nuclei or other conditions.

Table 4: Recommendations, valid for the conditions of table 3: TIP4P/Ice at 220 K (Ih/Ic also at 250 K) with five independent preparations; TIP4P/2005 at 220 and 200 K and mW at 220 K with three; six high-pressure ices and liquids at 0.3–3.0 GPa and 220 K with two; default neighbor settings unless stated. Rankings are the same for oracle and held-out scores. In these data, separations below about 0.65 did not reproduce across preparations.

Question	Best in the benchmark	Avoid
Ice vs. liquid, ambient	$\bar{s}$, $\bar{q}_6$, $\bar{Q}$; six-ring fraction if no averaging is wanted	CN, N_{HB}
Ih vs. Ic	$\bar{q}_3$, $\bar{q}_4$; F_4 if hydrogens are available	ring statistics, unaveraged parameters
Polytype labels in nuclei	CHILL+ (Ih, stacking-disordered), CHILL or CHILL+ (Ic); dead layer 3.4 Å (CHILL+), 5.0 Å (PTM) on basal faces at 220 K (10% thinner on prism faces), so labeled sizes of seeds of 200–1600 molecules are 33–65% (CHILL+) of the constructed size; recover sizes with eq. (9)	CHILL for Ih; comparing nucleus sizes from different labelers without correction
Nucleus detection in liquid	$\bar{s}$, $\bar{q}_6$, CHILL+, PTM (largest false cluster $\leq$8 molecules at 220 K)	smooth CN, d_5, $\bar{q}_4$; CHILL+ interfacial class as ice; crystalline connections and $\bar{s}$ in dense liquid
Distinguishing crystal phases	$\bar{s}$, smooth CN, Voronoi asphericity, $\bar{Q}$; $\bar{q}_6$ for VII/VIII	$\hat{w}_4$, $\hat{w}_6$, D, N_{HB}, A_3
High-pressure ice vs. liquid at the same pressure	per pair (fig. 8): $\bar{s}$ for III, VI, VII, VIII; six-ring fraction for II; for V, smooth CN with $\bar{q}_4$ (0.92; best single 0.85); no single parameter above 0.69 for all	ambient-pressure choices without checking
Proton-ordered vs. disordered	no local parameter tested is reliable (VII/VIII separate through their oxygen lattices)	–
Oxygen-only models (mW)	$\bar{s}$, $\bar{q}_6$, $\bar{Q}$, $\bar{q}_3$ (ice vs. liquid) and $\bar{q}_3$, $\bar{q}_4$ (Ih vs. Ic); their TIP4P/Ice thresholds transferred within 0.01 at 220 K, thermal frames, default settings	thresholds of distance parameters from other models

6.1 Software

Most parameters are available in more than one package, but with different conventions. freud [159] and pyscal [160] compute Steinhardt parameters (including averaged forms) and Voronoi tessellations; OVITO [161] provides PTM, identify-diamond, centrosymmetry and CHILL+; PLUMED [162, 163] provides differentiable versions of coordination and Steinhardt-type parameters, and with mlcolvar [144] learned CVs; MDAnalysis [164, 165] handles trajectories and hydrogen-bond analysis; DScribe [127] computes high-dimensional descriptors. SolOrder, used here, implements all parameters of table 2 on MDAnalysis universes with parallel, Numba-compiled kernels, and wraps SOAP, ACSF and MBTR through DScribe. Its thread parallelism pays off only for large systems and a few parameters (section 5.12). It is an analysis library: none of its parameters provides gradients, so CVs should be taken from PLUMED. Packages differ in neighbor rules, normalization and the averaging of q_{lm} vectors versus scalars; agreement between them should be checked, not assumed.

6.2 Reproducibility checklist

When reporting results that depend on a local order parameter, state:

1. the exact definition and the code and version used;
2. the neighbor rule (cutoff, number of neighbors or Voronoi), the averaging neighborhood, and whether hydrogens are used;
3. the water model and state point, and where that state point lies on the model's own phase diagram;
4. the threshold, how it was calibrated (on which reference populations, thermal or quenched), and whether it was tested on held-out data;
5. for labelers, the classes reported (including interfacial classes) and how clusters are built from labels;
6. for nucleus sizes, the observable counted (correct polytype, largest cluster, with or without interfacial classes), the size definition (equimolar or other), the labeler's dead layer or another correction for the surface layer it misses, and the largest false cluster it reports in liquid at the same state point;
7. for CVs, how continuity and gradients were checked.

7 Outlook

Four problems stand out. First, proton order. No local parameter we tested separates proton-ordered from proton-disordered forms of one oxygen framework, and those that seem to do so use the accompanying lattice distortion. Polarization and longer-range orientational correlations remain to be tested as local observables. Second, interfaces. Every labeler we tested needs several Ångströms of crystal before it assigns a molecule its bulk polytype, and labelers differ in what they call the remaining near-surface molecules, which makes the size of small nuclei and the width of interfaces depend on the labeler. Run F shows that, for spherical seeds of 200–1600 molecules in TIP4P/Ice at 220 K, this bias is predictable from a bulk recall and a dead layer per labeler, measured on a flat interface, and can be inverted to recover the size, also for seeds prepared after the calibration had been fixed. The dead layer is about 10% thinner on prism than on basal faces and nearly constant over 500 ps after release. The open questions are how it changes with temperature, nucleus shape and the direction of interface motion, how it behaves for learned classifiers, and how the corrected sizes relate to thermodynamically defined critical nuclei; the constructed seeds of run F are suited to the first of these. Third, transferability. Thresholds of averaged and bond-orientational parameters move between water models; thresholds of distance-based parameters do not, and learned models trained on one model should be tested on others, including machine-learned potentials of first-principles quality. Fourth, collective variables. Good per-molecule separation does not make a good CV; parameters need smooth neighbor membership and verified gradients, and their quality as reaction coordinates should

be judged by committor analysis rather than by separation of the end states. Amorphous ices, dense liquids at high pressure and confined water are the natural next datasets.

Data and code availability

SolOrder (version 1.1.0, MIT license), the benchmark scripts, settings registry and result tables are available in the SolOrder repository. The figures in section 5 are generated by `benchmarks/figures.py` from the CSV files in `benchmarks/results`. The archival dataset (`scripts/build_release.py`) contains all coordinates and cells, preparation metadata, the reference labels and their meaning, the fixed evaluation splits and every result table, with SHA-256 checksums; `examples/benchmark_quickstart.py` reproduces one published score from the coordinates in a few seconds. The confirmation protocol of run G, with the checksums of the frozen calibration, is `benchmarks/confirmation_protocol.md`, and the package versions of the analysis and simulation environments are recorded in the archive. The unbinding trajectories of the supplementary applications belong to separate path-sampling campaigns and are not part of the archive; the archive holds their per-water order parameters, region labels and splits, from which `host_guest.py` and `confined_water.py` reproduce every supplementary table and figure. [DOI after deposition.]

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